

K-FIELD

SPECTROPHOTOMETER pic_id14E

ANALYSIS

EXPOSED PLANE STRUCTURE

Full-valid, full-signal rectilinear embedding, reciprocal-axis families, complex spectral phase, nonlinear modulation, and the dimensionally recovered transverse base face

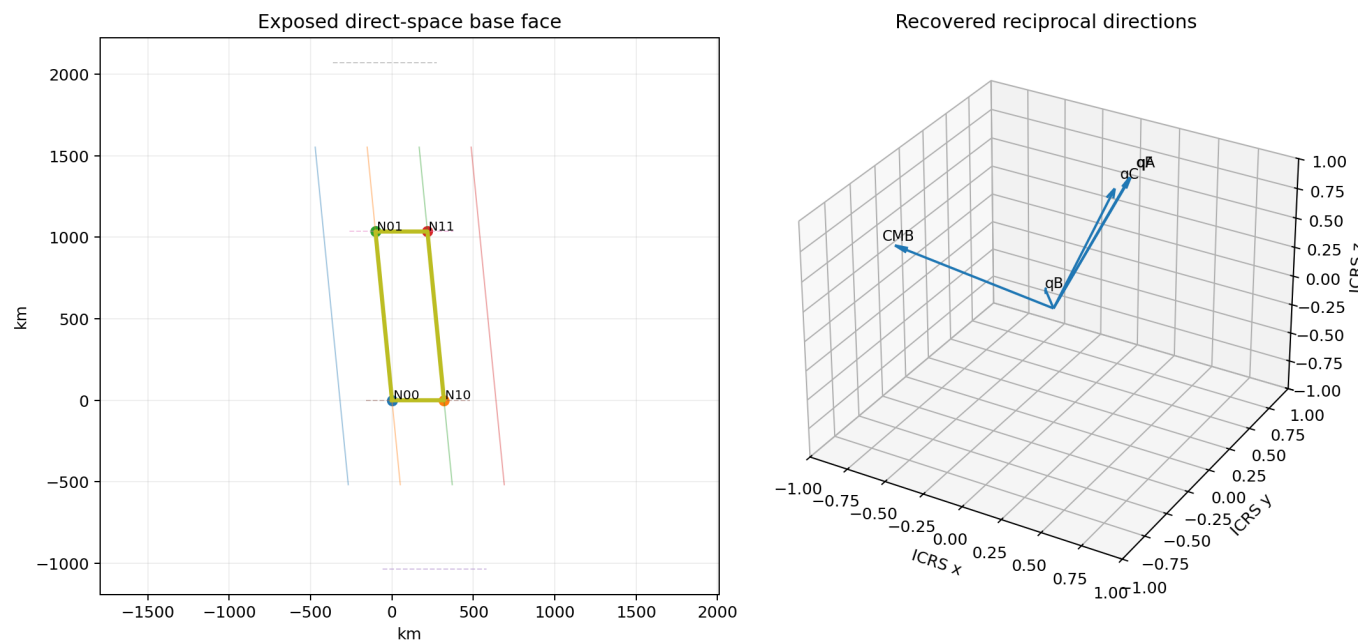


Figure 1. Data-derived title composite. Left: the dimensionally recovered transverse direct-space base face bounded by the qA and qB plane families. Right: the measured reciprocal directions qA, qB, qC, qF, and the supplied CMB axis in ICRS coordinates.

Principal Investigator
P. Dwayne Esterline
BS, Huntington University (1999)
MBA, Indiana Wesleyan University (2008)
Krytronx, LLC.

PROPRIETARY CONFIDENTIAL INFORMATION AND TRADE SECRET OF KRYTRONX, LLC. ALL RIGHTS RESERVED. DO NOT DISTRIBUTE.

Archival publication epoch: 1785932820 | Generated UTC: 2026-08-05T12:27:00Z
Machine-readable companion: K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane_structure_1785932820.json
Rendered with Python ReportLab using Helvetica, Helvetica-Bold, Times-Roman, and Courier only.

Abstract

This archival reference presents a self-contained forensic signal analysis of the 24.064-hour pic_id14E rotating spectrophotometer record. The source contains 169,555 measurements, 1,883 complete 90-position revolutions, and a terminal 85-row partial revolution. The analysis uses every valid sample within explicitly reconstructed channel-freshness apertures. It does not use a holdout partition and does not convert numerical rankings into proof claims. The detector trajectory is correctly unwrapped as the sum of the supplied CMB translation, Earth rotation, laboratory position, and measured device rotation. Repeated local orientations are therefore recognized as distinct inertial positions rather than collapsed into closed laboratory circles.

The full-valid signal contains recurrent CMB-transverse reciprocal-axis families rather than one isolated plane spacing. Two independently extracted directions, qA and qB, are separated by 84.391 degrees; their cross product is 0.0364 degrees from the supplied CMB axis. The pair exposes a dimensionally recoverable transverse direct-space base face. Its direct vectors have lengths 319.393 km and 1,039.778 km, an included angle of 95.609 degrees, and an area of 330,508.374 square kilometres. The reciprocal plane spacings are 317.864 km and 1,034.800 km. The longitudinal direct-space period is not resolved by this trajectory, so the measured object is a transverse base face and two plane families rather than a closed three-dimensional primitive cell.

Information is distributed across broadband and spectral-polarization principal components, complex channel phase, quadrature loops, nontrivial phase locks, carrier-envelope sidebands, within-turn angular graphs, and low-order sum/difference phases. A local reciprocal relation qC approximately equals qA plus 2 qB with 1.303 percent vector mismatch. This relation organizes direction and magnitude locally but does not establish global phase registry because its small CMB-parallel residual accumulates approximately 146 cycles across the full translation. The resulting paradigm is a multiplex rectilinear embedding: two transverse reciprocal directions organize a multidimensional spectral state while the unresolved longitudinal coordinate follows the dominant translation axis.

Core result. The entire valid record exposes a two-dimensional CMB-transverse reciprocal frame embedded in three-dimensional inertial motion. The recovered frame organizes multiscale spectral polarization, nonlinear phase relationships, geometric derivatives, and mixed-axis modulation. It establishes a measurable transverse plane structure and direct-space base face, while leaving the longitudinal period and absolute three-dimensional lattice registry unresolved.

Table of Contents

Abstract	2
1. Purpose, Scope, and Analytical Posture	6
1.1 What is measured and what is inferred	6
1.2 Full-signal exploratory convention	6
2. Experimental Record and Forensic Data Integrity	7
2.1 Effective timing and turn reconstruction	7
2.2 Channel freshness regimes	7
3. Correctly Unwrapped Three-Dimensional Motion	9
3.1 Position equation	9
3.2 Supplied translation vector and laboratory geometry	9
3.3 Consequence of the scale separation	10
4. Signal Construction and Exhaustive Decomposition	11
4.1 Per-turn temporal observables	11
4.2 Angle-resolved profiles	11
4.3 Multiscale temporal bands	11
4.4 Complex phase and nonlinear relations	11
4.5 Rectilinear path search	11
5. Full-Valid Multiscale State Dimension	12
5.1 Initial ten-channel spectral modes	13
6. Recurrent Reciprocal Axes	15
6.1 Orthogonal-frame closure	15
6.2 Axis-family clustering and spacing branches	16
7. Exposed Direct-Space Base Face and Plane Families	17
7.1 Dual-basis derivation	17
7.2 Base-face node ledger	18
7.3 Plane equations and three-dimensional intersections	18
8. Hidden Geometric Derivatives	20

8.1 Sidereal velocity form	20
8.2 Spatial phase traversal	21
8.3 qC projected-velocity reversals	21
9. Full-Signal Rectilinear Path Rankings	23
9.1 Relationship to the earlier 887 km and 1,566 km paths	24
10. Two-Axis Rectilinear Modulation and Torus Embedding	25
10.1 Low-order reciprocal lattice basis	25
10.2 Two-dimensional phase maps	26
11. Low-Order Reciprocal-Vector Grammar	31
12. Complex Spectral Phase and Hidden Channel Organization	33
12.1 Antipodal and quadrature edges	34
12.2 Continuous phase rotation across scale	34
13. Hidden Nonlinear Relations and Quadrature Loops	35
13.1 Interpretation of quadrature coordinates	37
14. Nontrivial Phase Locks and Late-State Frequency Division	38
15. Sidebands, Carrier Envelopes, and Phase-Amplitude Coupling	40
15.1 Phase-amplitude coupling	42
16. Angular Harmonic Graph as a Separate Multiplex Layer	43
16.1 Direct rectilinear representation of within-turn harmonics	43
17. Regime Continuity and Recovery	45
18. Integrated Geometric Interpretation	46
18.1 Minimum sufficient structural model	46
18.2 What the exposed base face means	46
18.3 Why small vector errors become large phase errors	46
18.4 Paradigm consequence	46
19. Reproduction Procedure	47
19.1 Defensive archival requirements	47
20. Conclusions	48

Appendix A. Complete Exposed Base-Face Node Patch 49

 Appendix A.1 Plane-family ledger 49

Appendix B. Axis-Family Branch Assignments 50

Appendix C. Synchronized Channel Phase Ledger 51

Appendix D. Spectral Phase Polynomial Fits 53

Appendix E. Archival and Reproduction Record 54

 Appendix E.1 Large-ledger manifest 54

 Appendix E.2 Figure manifest 55

1. Purpose, Scope, and Analytical Posture

The objective is to expose the spatial organization carried by the correctly unwrapped multichannel signal. The analysis is exploratory: it reports observed structure, derived geometry, recurrence, model compression, and unresolved degrees of freedom. Numerical fit values are used to compare representations and to identify where information is concentrated. They are not used as a gate that suppresses structures before they can be examined.

The record is treated as a graph with several coupled node types. Time samples, device angles, spectral channels, analytic phases, frequency ridges, reciprocal vectors, direct-space nodes, and plane crossings are all represented explicitly. Edges encode temporal coherence, complex phase, nonlinear dependence, harmonic matching, reciprocal-vector arithmetic, spatial-phase traversal, and geometric derivative alignment. The graph is multiplex because the within-turn angular relations and the slower translation-driven relations are not identical layers.

The analysis uses the full valid signal inside each channel-freshness aperture. It does not concatenate through stale-register intervals. This distinction permits exhaustive use of the available record without allowing a frozen channel to masquerade as a coherent spatial state.

1.1 What is measured and what is inferred

The empirical layer consists of channel counts, timestamps, device indices, channel freshness, per-turn profiles, per-turn means, moving standard deviations, and their spectra. The derived layer consists of inertial coordinates, analytic phase, principal components, complex coherence, path scores, reciprocal vectors, direct-space dual vectors, and derivative events. The structural inference is that several of these derived quantities recur in a common transverse geometric frame. The unresolved question is whether a unique longitudinal period exists and, if so, how it should be measured.

1.2 Full-signal exploratory convention

All valid samples are used within each aperture.

No holdout partition is applied.

No null-based acceptance rule is used to remove candidate structure.

Rankings, recurrence, closure, and compression are reported as observed properties.

2. Experimental Record and Forensic Data Integrity

The source record spans slightly more than one sidereal day of laboratory motion. The device performs 90 indexed measurements per revolution, with index zero directed down. Increasing index sweeps toward north, then up, south, and back down in a vertical plane about an east-west axis. Each local circle is embedded in a translating and Earth-rotating laboratory trajectory.

The raw record contains partial sensor failures in which subsets of spectral registers stop changing while other channels remain active. These events must be treated as changes in measurement rank. They do not stop the timing stream or the entire instrument, but they change which multichannel state spaces are observable.

Field	Value
Source file	pic_id14E_full24_hrs_20260805.csv
SHA-256	6f5e3c69c623fe02b854e121110f915861bab24e8b6f000b094e810f49b53f62
Measurements	169,555
Complete 90-position turns	1,883
Terminal partial revolution	85 rows
First effective UTC	2026-08-04T10:55:12.483845+00:00
Last effective UTC	2026-08-05T10:59:04.362935+00:00
Duration	24.064410858 h
Earth rotation	361.954456 deg
Median revolution period	45.984987 s
Rows with health_flag = 0	269

2.1 Effective timing and turn reconstruction

```
t_effective = t_date.UTC - frame_read_us
```

Sequential revolutions are reconstructed from index-zero resets because the source photo counter wraps. A revolution is retained only when it contains exactly 90 rows whose indices are the ordered sequence 0 through 89. This produces 1,883 complete turns. The terminal 85-row partial revolution is preserved in the source archive but excluded from complete-turn profile analysis.

2.2 Channel freshness regimes

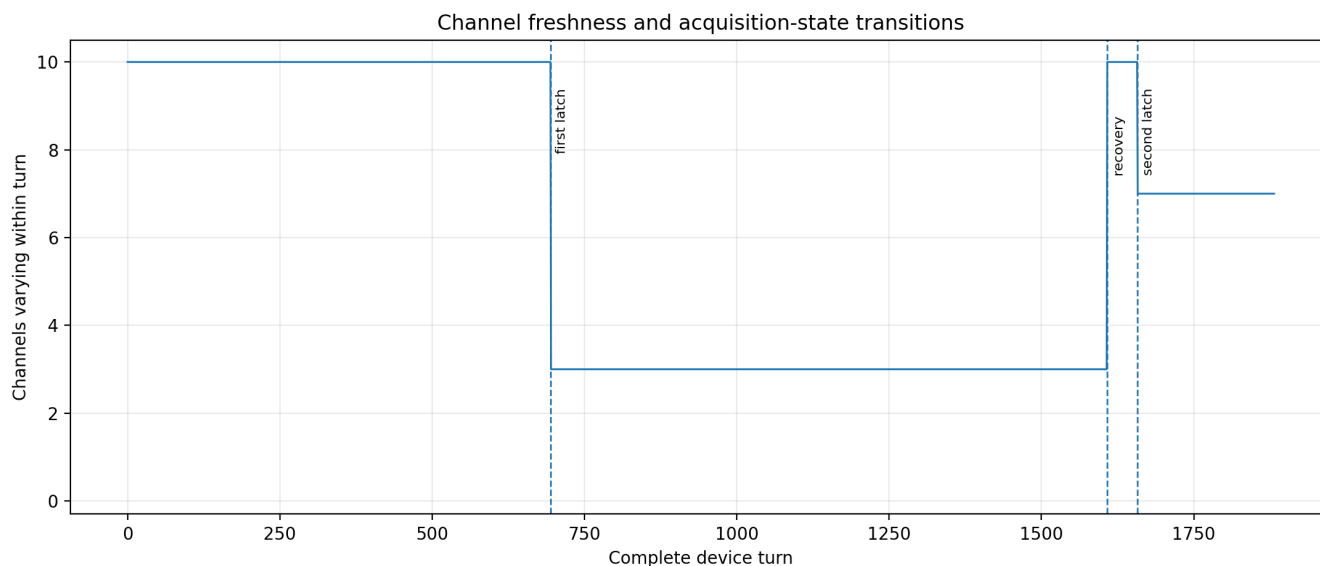


Figure 2. Channel freshness across the full record. The measurement rank changes twice. The first latch leaves 445 nm, Clear, and NIR active; the second latch leaves the red-visible channels plus Clear and NIR active. All subsequent analyses use only the channels fresh in the named aperture.

Aperture	Start turn	End turn	Turns	Duration h	Fresh channels
A_full10	0	693	694	8.852018	415;445;480;515;555;590;630;680;clear;NIR

Aperture	Start turn	End turn	Turns	Duration h	Fresh channels
Bridge3	0	1656	1,657	21.152904	445;clear;NIR
C_recovery10	1609	1656	48	0.600357	415;445;480;515;555;590;630;680;clear;NIR
D_late7	1657	1882	226	2.874036	515;555;590;630;680;clear;NIR

The initial 694-turn interval is the principal ten-channel discovery aperture. The 1,657-turn bridge supplies a long continuous record for 445 nm, Clear, and NIR. The 48-turn recovery interval offers a short return to all ten channels. The final 226-turn interval supports seven red-visible/NIR channels. These apertures overlap by design because they answer different questions with different valid channel sets.

3. Correctly Unwrapped Three-Dimensional Motion

A laboratory rotation is closed only in the local apparatus frame. In inertial space, the detector is translated by the supplied CMB vector while Earth rotates and while the detector moves around its east-west axle. The correct trajectory is therefore open and three-dimensional.

This distinction is fundamental. If each revolution were collapsed onto the same local circle, spatial phase accumulated during translation would be erased. The unwrapped model retains both orientation and position.

3.1 Position equation

$$\mathbf{R}(t, \theta) = \mathbf{g} (t - t_0) + \mathbf{r}_{\text{lab, ICRS}}(t) + r_d \mathbf{u}_{\text{ICRS}}(t, \theta)$$

Here $\mathbf{g}$ is the supplied CMB velocity vector, $\mathbf{r}_{\text{lab, ICRS}}$ is the Earth-rotating laboratory position, r_d is the detector radius from the rotation axis, and $\mathbf{u}_{\text{ICRS}}$ is the instantaneous detector direction. Coordinates are reported relative to a convenient origin; subtracting an origin does not close the trajectory.

3.2 Supplied translation vector and laboratory geometry

Quantity	Value
CMB velocity x	-880,221.981635 m/s
CMB velocity y	-143,911.436562 m/s
CMB velocity z	+266,335.026949 m/s
CMB unit vector	(-0.945636157362, -0.154606293310, +0.286127859460)
Laboratory latitude	42.129 deg N
Laboratory longitude	83.144 deg W
Laboratory altitude	213.36 m
Device radius	0.041529 m

Correctly unwrapped 24-hour laboratory trajectory

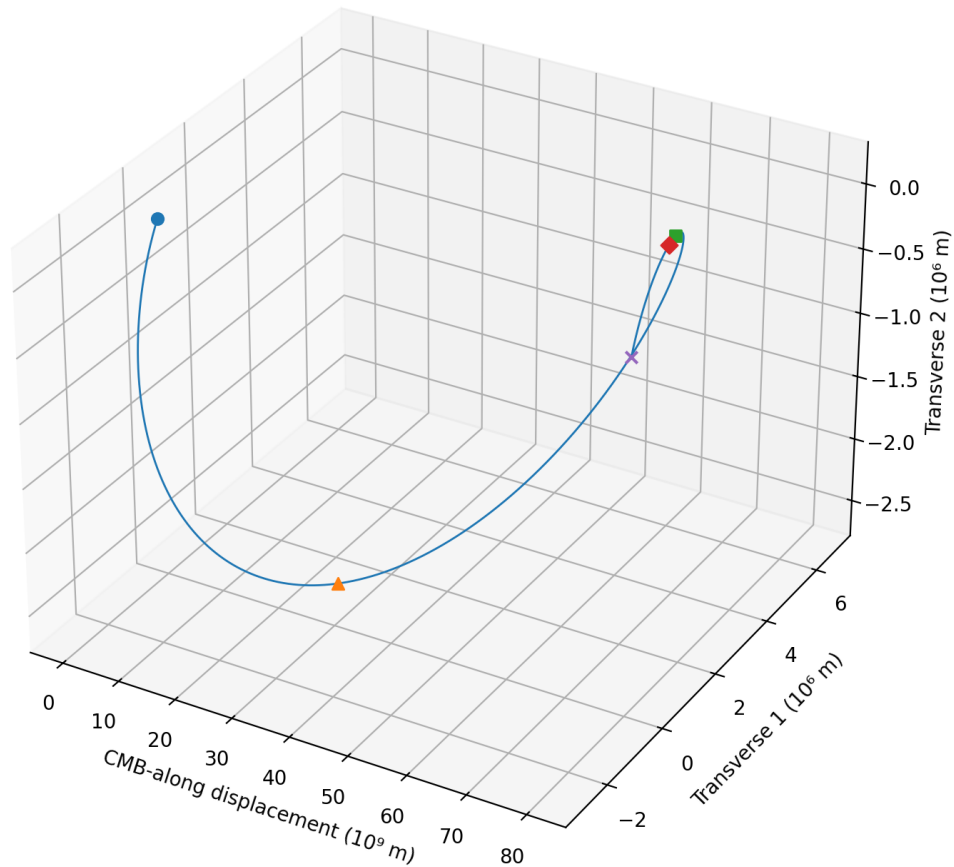


Figure 3. Correctly unwrapped trajectory in a CMB-aligned basis. The along-track motion is tens of billions of metres, while Earth rotation produces transverse excursions of order millions of metres. The visible curve is not a stack of closed circles; it is a translating, rotating path through new spatial coordinates.

3.3 Consequence of the scale separation

A tens-of-turn modulation cannot have a large reciprocal-vector component along the CMB velocity. A substantial parallel component would be crossed at approximately 930.8 km/s and would produce a much shorter period. The recovered axes are therefore tightly concentrated in the CMB-transverse plane. This geometric restriction emerges repeatedly from independent signal observables.

4. Signal Construction and Exhaustive Decomposition

The analysis preserves both the temporal state of each revolution and the angle-resolved structure within each revolution. No single representation is assumed to contain all information. A channel can be weak in a raw mean yet strong in a phase relation, a moving-variance harmonic, a nonlinear loop, or a mixed spatial phase.

The main representations are listed below so that the subsequent geometric results can be read as transformations of explicit measured quantities.

4.1 Per-turn temporal observables

For each complete revolution and channel, the arithmetic mean, logarithmic mean, range, and standard deviation are computed. The logarithmic mean converts multiplicative drift into an approximately additive coordinate and makes inter-channel state decomposition more stable.

$$L_c(k) = \log(\text{mean_theta } x_c(k, \text{theta}))$$

4.2 Angle-resolved profiles

The raw 90-position profile is detrended within each turn by removing its fitted intercept and linear index trend. A separate circular 31-position moving standard deviation is computed at every device angle. The latter emphasizes local texture and modulation density rather than mean intensity. Complex Fourier harmonics $m = 1$ through 8 are retained for both observables.

$$C_{c,m}(k) = \text{mean_theta } [x_c(k, \text{theta}) \exp(-i m \text{theta})]$$

4.3 Multiscale temporal bands

Zero-phase band decomposition separates 3-12, 12-64, 64-300, and, where the aperture is long enough, 300-1000 turn structures. These bands are not arbitrary smoothing labels. They isolate the rapid common mode, the principal spectral-polarization region, slow nonlinear loops, and ultraslow antipodal organization.

4.4 Complex phase and nonlinear relations

The analytic signal from the Hilbert transform supplies instantaneous amplitude and phase. Complex coherence retains phase even when ordinary Pearson correlation approaches zero. Quadrature regression represents one channel by both the in-phase and quadrature coordinates of another. Mutual information and distance correlation expose curved dependencies. Nontrivial $m:n$ phase locks detect frequency division and harmonic phase organization.

$$\begin{aligned} z_c(t) &= x_c(t) + i H\{x_c(t)\} \\ \text{coherence_ij} &= | \text{mean}(z_i \text{ conj}(z_j)) | / \sqrt{\text{mean}|z_i|^2 \text{ mean}|z_j|^2} \\ \text{phase_ij} &= \arg \text{mean}(z_i \text{ conj}(z_j)) \end{aligned}$$

4.5 Rectilinear path search

A fixed reciprocal vector q predicts an instantaneous crossing frequency from the projected inertial velocity. The time-frequency search therefore follows geometry-constrained curves rather than arbitrary polynomials in time.

$$\begin{aligned} \Phi_{q,t} &= q \cdot R(t) \\ f_{q,t} &= | d \Phi_{q,t} / dt | = | q \cdot v(t) | \\ P_{q,t} &= T_{\text{turn}}(t) / f_{q,t} \end{aligned}$$

5. Full-Valid Multiscale State Dimension

The multichannel state is low dimensional but not scalar. Its dimension changes with temporal scale. The rapid state is dominated by a common broadband coordinate; slower intervals separate into a common coordinate and one or more polarized coordinates.

Participation dimension is calculated from the PCA eigenvalue spectrum. A value near one means that one state coordinate dominates. A value near two means that two coordinates carry most of the variance even if many detector channels are present.

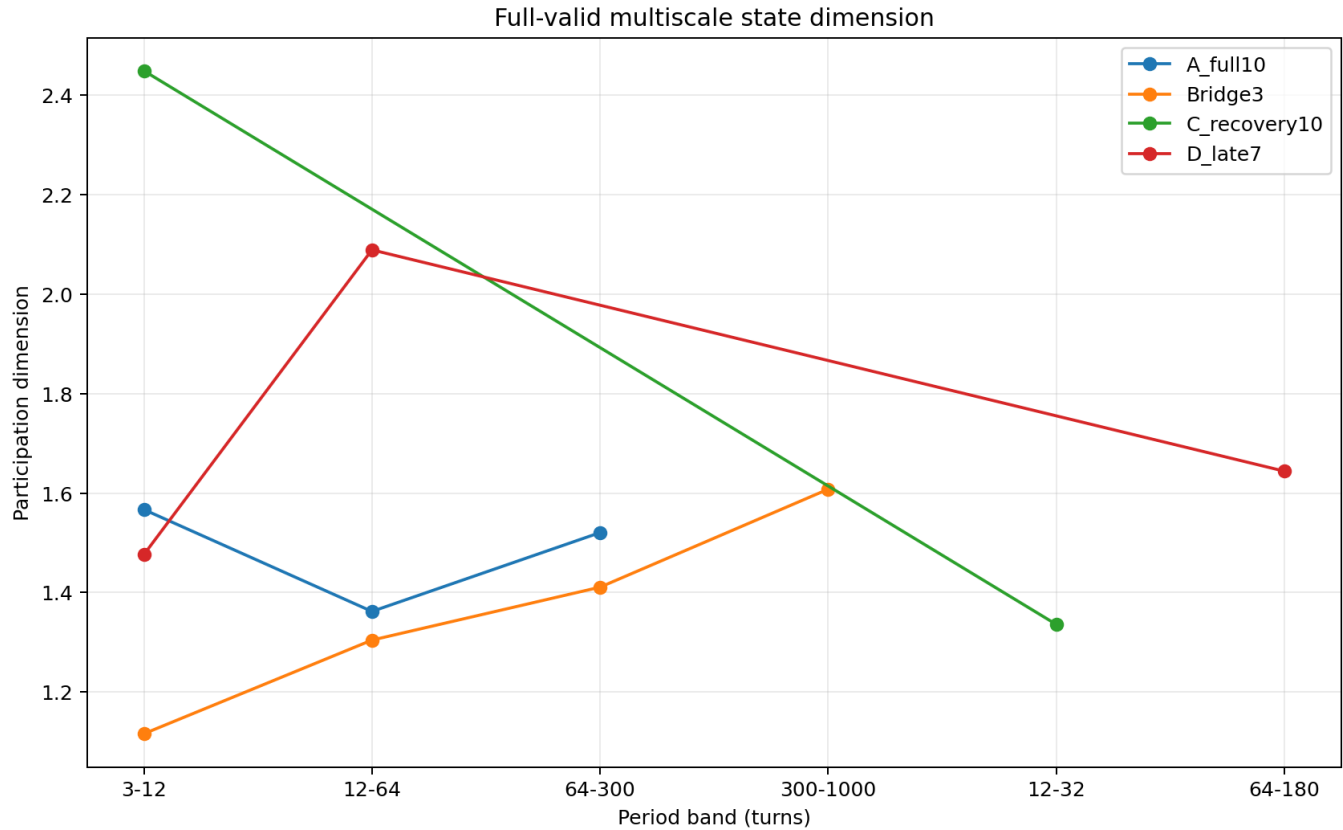


Figure 4. Participation dimension across every valid aperture and supported period band. The initial ten-channel state expands from a mostly one-dimensional rapid mode toward an approximately two-dimensional slow state. The long three-channel bridge also develops a distinct ultraslow organization.

Aperture	Band	Turns	PC1	PC2	PC3	Participation dimension
A_full10	3_12	694	79.18%	8.59%	4.93%	1.567
A_full10	12_64	694	85.40%	5.01%	3.35%	1.362
A_full10	64_300	694	79.93%	12.82%	4.41%	1.521
Bridge3	3_12	1657	94.56%	3.91%	1.53%	1.116
Bridge3	12_64	1657	86.97%	9.46%	3.57%	1.304
Bridge3	64_300	1657	83.04%	13.38%	3.58%	1.411
Bridge3	300_1000	1657	75.28%	23.48%	1.24%	1.608
C_recovery10	3_12	48	60.93%	14.21%	10.25%	2.449
C_recovery10	12_32	48	85.73%	11.43%	2.02%	1.336
D_late7	3_12	226	81.87%	5.79%	3.98%	1.476
D_late7	12_64	226	65.05%	22.57%	5.51%	2.089
D_late7	64_180	226	74.14%	24.15%	1.57%	1.644

5.1 Initial ten-channel spectral modes

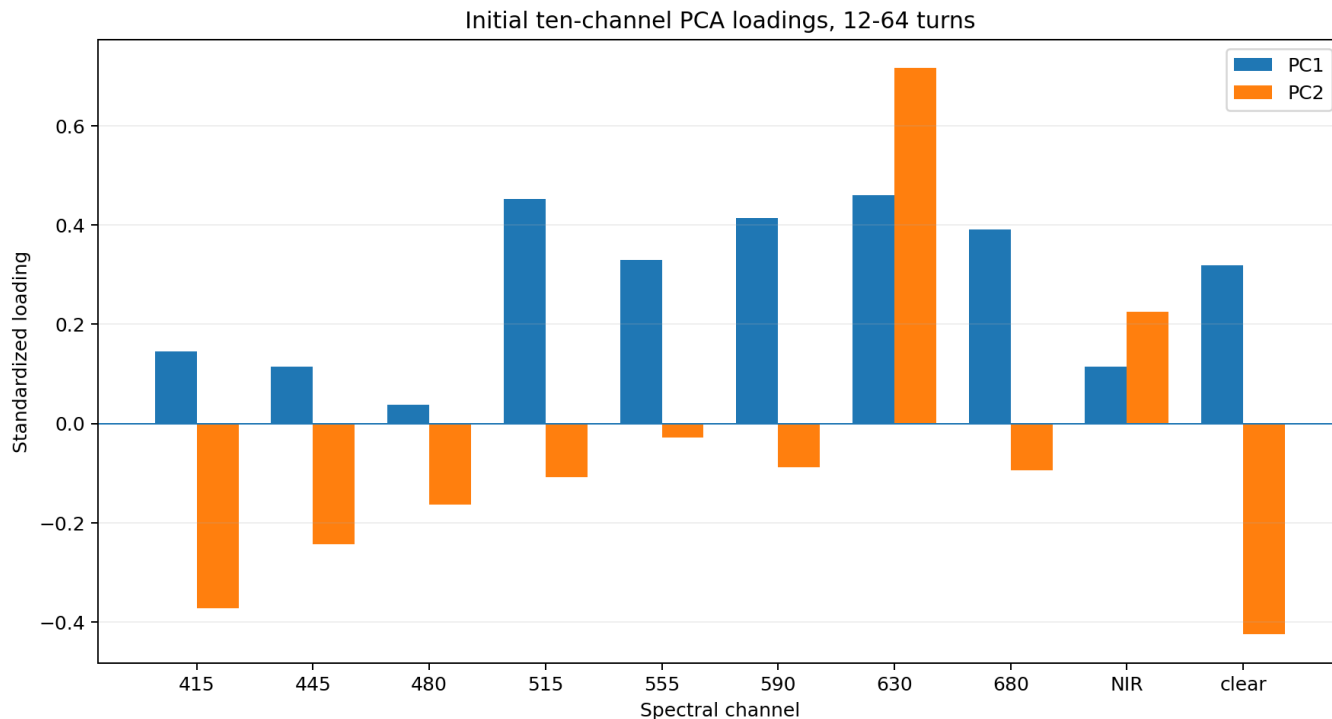


Figure 5. Initial ten-channel loadings in the 12-64-turn band. PC1 is the broadband coordinate. PC2 separates the short-wavelength blue channels from the red-visible and NIR response, exposing a spectral-polarization axis.

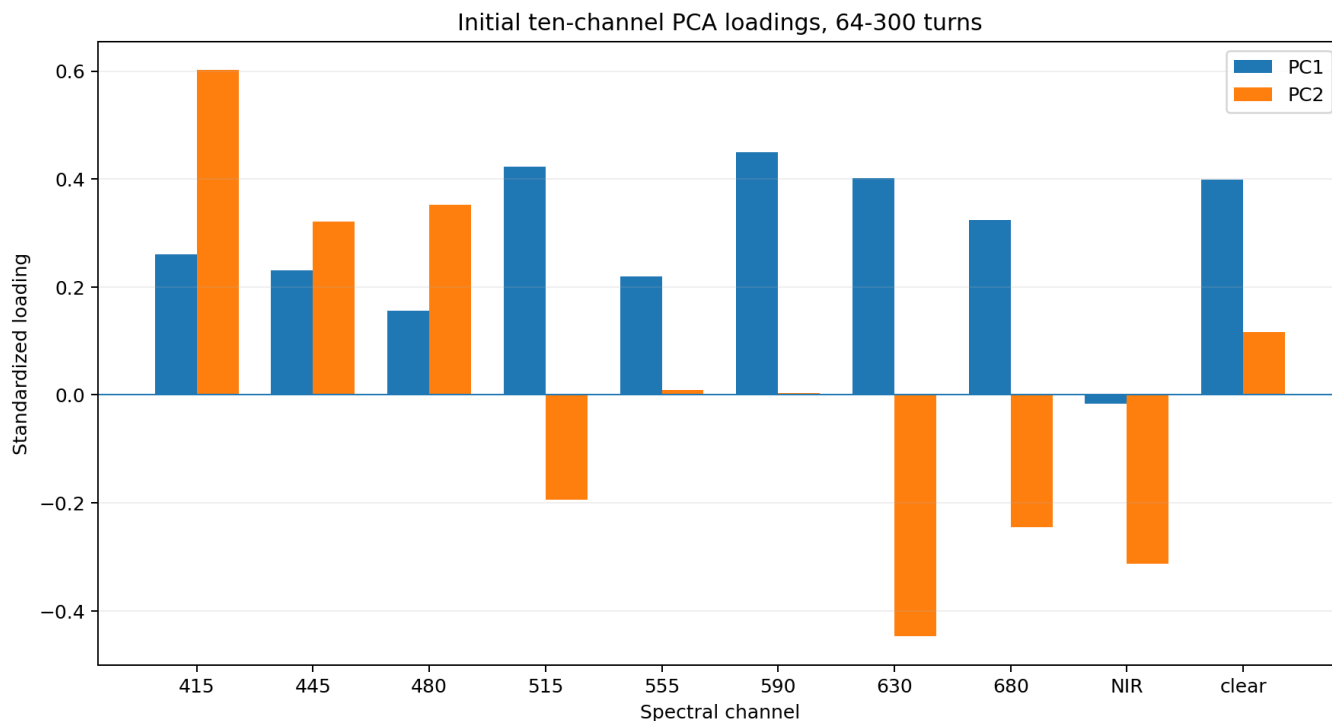


Figure 6. Initial ten-channel loadings in the 64-300-turn band. The second mode becomes a larger fraction of the state, consistent with the increasing participation dimension and the strengthening of complex phase organization.

The spectral PC2 loading is ordered by photon energy and changes sign near the yellow-green region. This behavior is consistent with a small displacement or tilt of the measured spectral distribution: a shift in spectrum produces a response proportional to the energy derivative of the mean spectrum. The important point is not a specific source mechanism; it is that multiple detector channels encode a common latent coordinate with different signs and phases.

$$I(E - \delta E) \approx I(E) - \delta E \frac{dI}{dE}$$

6. Recurrent Reciprocal Axes

Full-signal searches do not isolate one universal spacing. They reveal recurrent directions populated on several spacing branches. Four representative axes are especially useful because they originate in different observables and organize different layers of the signal.

qA is extracted from the clean spectral-polarization PC2. qB is extracted from the first principal component of the 31-position moving-standard-deviation angular harmonic state. qC is extracted from the clean broadband PC1. qF is the strongest full-bridge differential carrier.

Axis	Originating observable	Spacing	RA	Dec	Angle to CMB
qF	1,657-turn bridge differential	670.886 km	79.653 deg	+48.332 deg	89.98362 deg
qA	Clean ten-channel spectral PC2	317.864 km	81.381 deg	+45.796 deg	89.98418 deg
qB	Clean 31-position variance PC1	1034.800 km	296.484 deg	+44.634 deg	89.96584 deg
qC	Clean ten-channel broadband PC1	260.277 km	46.645 deg	+69.405 deg	89.99683 deg

Exploratory reciprocal-frame directions

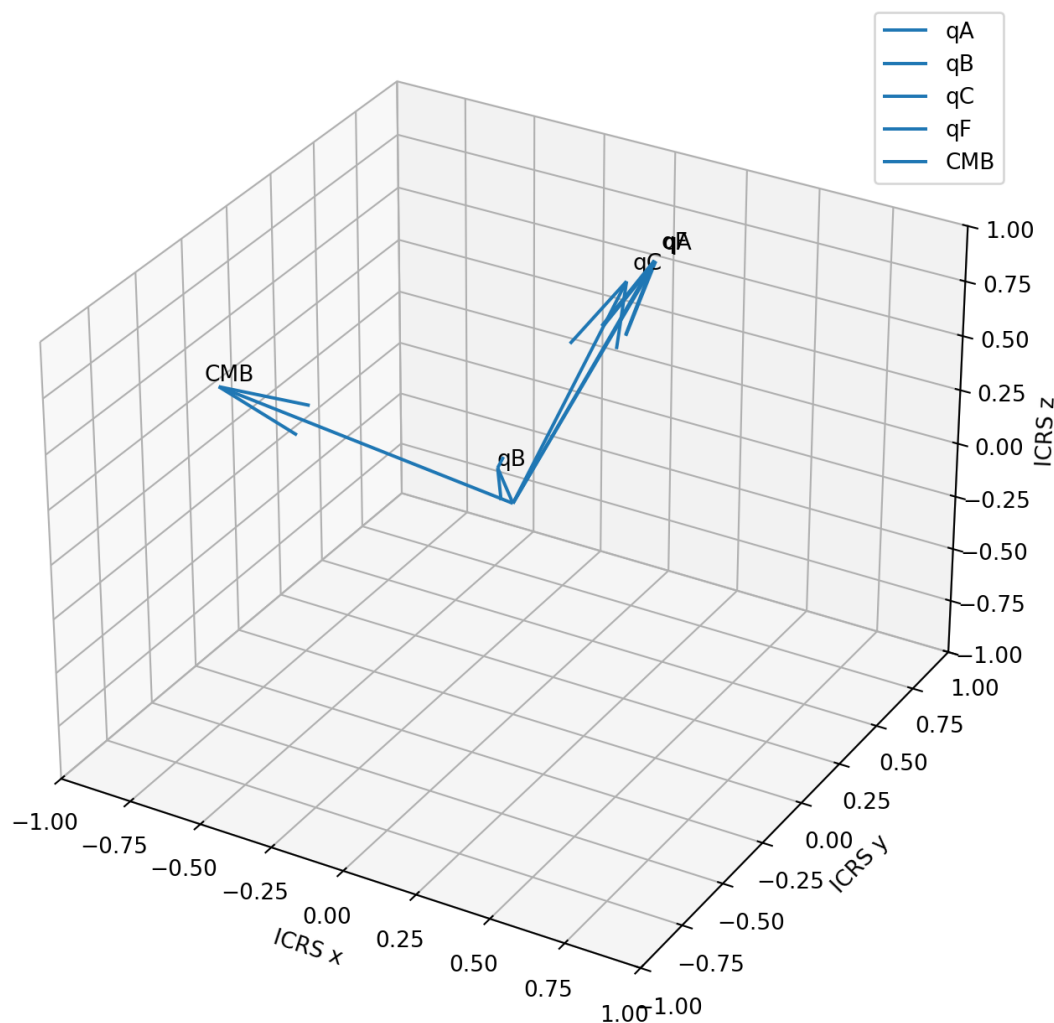


Figure 7. Recovered reciprocal directions in ICRS unit-vector coordinates. qA, qC, and qF occupy one broad transverse sector; qB occupies a second sector. The supplied CMB axis is nearly normal to the plane spanned by qA and qB.

6.1 Orthogonal-frame closure

qA and qB are separated by 84.391343 degrees. Their cross product is 0.036386 degrees from the supplied CMB axis. This is a three-dimensional directional closure derived from two independently extracted signal coordinates. The sign of the cross product is axial; reversing it describes the same geometric line.

```
c_hat = (qA x qB) / |qA x qB|
angle_axis(c_hat, e_CMB) = 0.036386 deg
```

6.2 Axis-family clustering and spacing branches

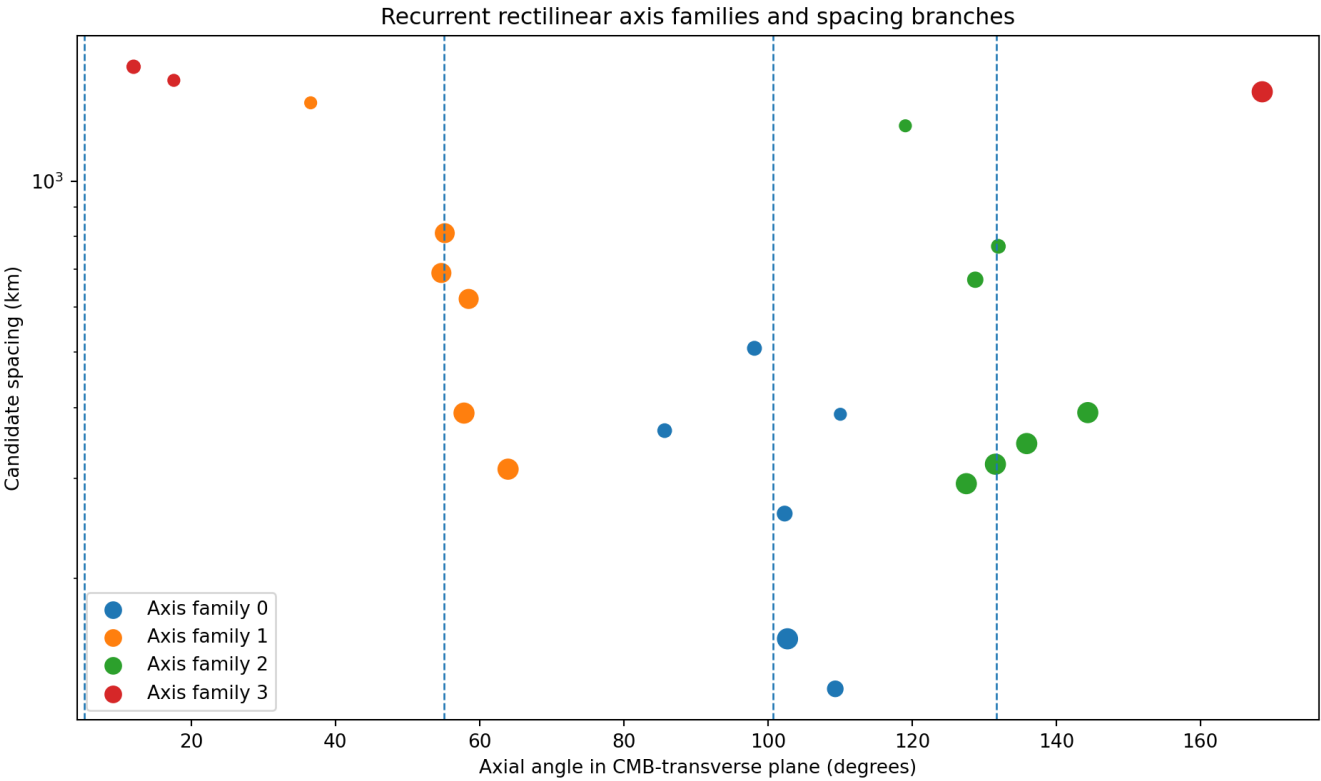


Figure 8. Recurrent axis families and spacing branches from the full-signal candidate bank. Directional recurrence is stronger than any claim of one exact spacing ladder: independent observables repeatedly return to the same transverse sectors while occupying different wavelength-like branches in reciprocal magnitude.

Family	Members	Distinct observables	Transverse angle	RA	Dec	Candidate base spacing	Branch residual RMS
3.000000	3	1	5.145	100.800	-4.918	1513.413 km	0.0426
1.000000	6	4	55.080	301.574	+51.782	1511.612 km	0.2217
0	6	5	100.731	42.796	+70.289	1056.605 km	0.1898
2.000000	7	2	131.742	81.490	+45.634	1489.001 km	0.2189

The four family centres lie near transverse angles 5, 55, 101, and 132 degrees. Three families are populated by multiple independent observables. The candidate branch spacings range from approximately 128 km to 1.86 Mm. The branch residuals show that a harmonic interpretation is useful as an organizing grammar, but the record does not force one exact integer ladder.

7. Exposed Direct-Space Base Face and Plane Families

Reciprocal vectors describe plane normals and plane spacing. A direct-space base face is obtained by finding vectors a and b that advance one complete phase cycle in one plane family while advancing zero cycles in the other. Because only two independent transverse reciprocal vectors are resolved, the result is a two-dimensional base face embedded in three-dimensional ICRS space.

This construction is exact for the selected qA and qB vectors. It does not assume that the plane normals are perfectly orthogonal.

7.1 Dual-basis derivation

$$\begin{aligned} Q &= [qA^T ; qB^T] \\ D &= Q^T (Q Q^T)^{-1} = [a \ b] \\ Q D &= I_2 \end{aligned}$$

The minimum-norm columns of D lie in the span of qA and qB and satisfy $qA \cdot a = 1$, $qB \cdot a = 0$, $qA \cdot b = 0$, and $qB \cdot b = 1$. Each coefficient is in cycles per metre, so a and b are in metres.

Direct-space quantity	Measured value
a vector in ICRS	(23.580892, 241.209359, 208.024245) km
b vector in ICRS	(320.888448, -735.846440, 660.832416) km
a	319.393331 km
b	1039.778275 km
Angle(a,b)	95.608657 deg
Base-face area	330,508.374179 km ²
qA plane spacing	317.864281 km
qB plane spacing	1034.800487 km
Longitudinal period	Unresolved by the present trajectory

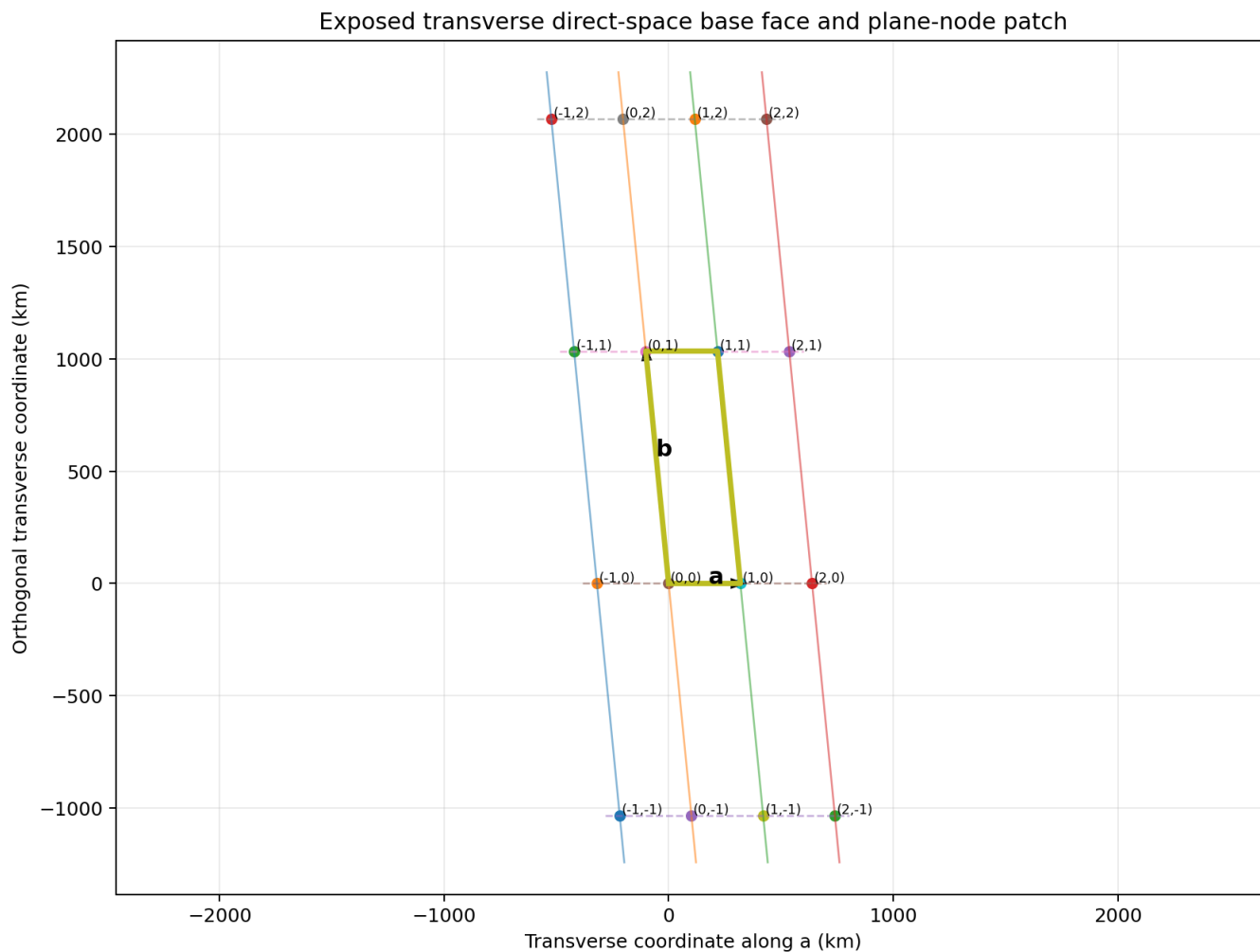


Figure 9. Dimensionally correct transverse base-face patch. Solid and dashed line families are the qA and qB phase planes in cross-section. Every displayed node is an integer intersection (i,j), where $qA \cdot r = i$ and $qB \cdot r = j$. The heavy parallelogram is the exposed primitive base face.

7.2 Base-face node ledger

Node	i	j	ICRS x km	ICRS y km	ICRS z km	qA phase	qB phase
N00	0	0	0.000000	0.000000	0.000000	0.000000	0.000000
N10	1	0	23.580892	241.209359	208.024245	1.000000	0.000000
N01	0	1	320.888448	-735.846440	660.832416	-0.000000	1.000000
N11	1	1	344.469340	-494.637081	868.856661	1.000000	1.000000

N00 is the chosen origin. N10 is one direct a translation, N01 is one direct b translation, and N11 is their sum. The node coordinates are not arbitrary drawing coordinates; they are the ICRS components obtained from the measured reciprocal vectors.

7.3 Plane equations and three-dimensional intersections

$$P_A(n): 3.287321093305e-07 \, x + 2.168679499253e-06 \, y + 2.255229490922e-06 \, z = n$$

$$P_B(m): 3.066748724203e-07 \, x + -6.155152953415e-07 \, y + 6.789419322445e-07 \, z = m$$

Each intersection of $P_A(n)$ and $P_B(m)$ is a line parallel to $qA \times qB$. The base-face nodes are the intersections of those lines with the chosen transverse cross-section. Without a measured longitudinal reciprocal vector, each node is physically an unresolved line class rather than a fully indexed three-dimensional lattice point.

Measured plane families P_A and P_B bounding the exposed base face

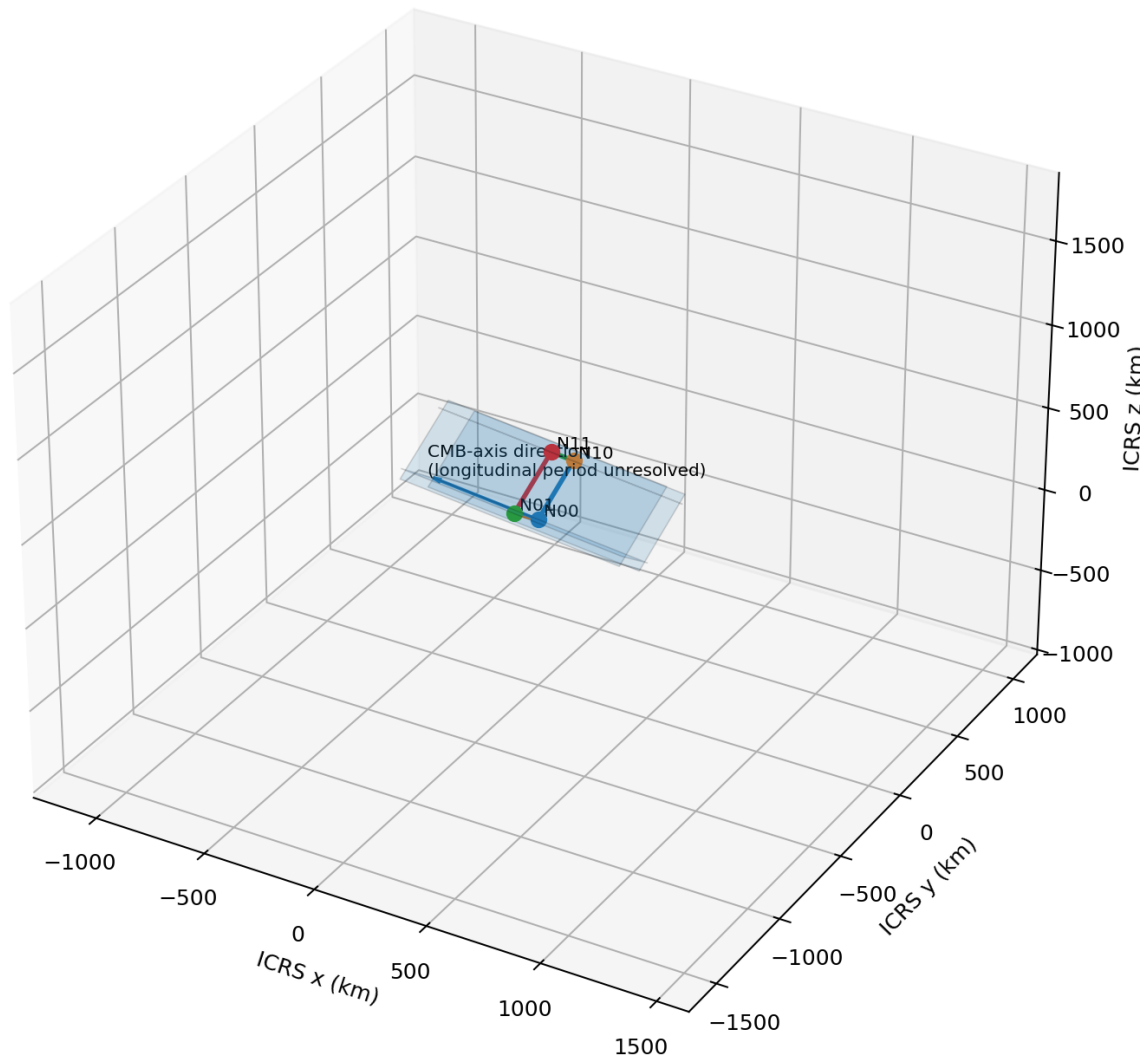


Figure 10. Three-dimensional plane families P_A(0), P_A(1), P_B(0), and P_B(1), with the four base-face nodes at the transverse section. The displayed extrusion along the CMB-axis direction is a visualization span derived from the measured base-face scale; it is not a claimed longitudinal period.

Structural boundary. The data expose two plane families and their direct-space base face. They do not yet expose the closing longitudinal pair of planes required to define a finite three-dimensional primitive cell.

8. Hidden Geometric Derivatives

A fixed reciprocal vector sampled by a rotating Earth has a constrained derivative hierarchy. Its projected speed is the sum of a constant translation term and one sidereal sinusoid. Acceleration and jerk are therefore not independent fitted curves; they are successive derivatives of the same geometry.

These derivatives provide a direct route from a time-frequency path to spatial structure. A frequency contraction, expansion, or reversal can be read as the projected laboratory velocity changing relative to a fixed plane normal.

8.1 Sidereal velocity form

$$\begin{aligned} v_q(t) &= v_0 + A \cos(\Theta(t)) + B \sin(\Theta(t)) \\ V_{\text{rot}} &= \sqrt{A^2 + B^2} \\ a_q(t) &= \omega_E [-A \sin(\Theta) + B \cos(\Theta)] \\ j_q(t) &= -\omega_E^2 [A \cos(\Theta) + B \sin(\Theta)] \end{aligned}$$

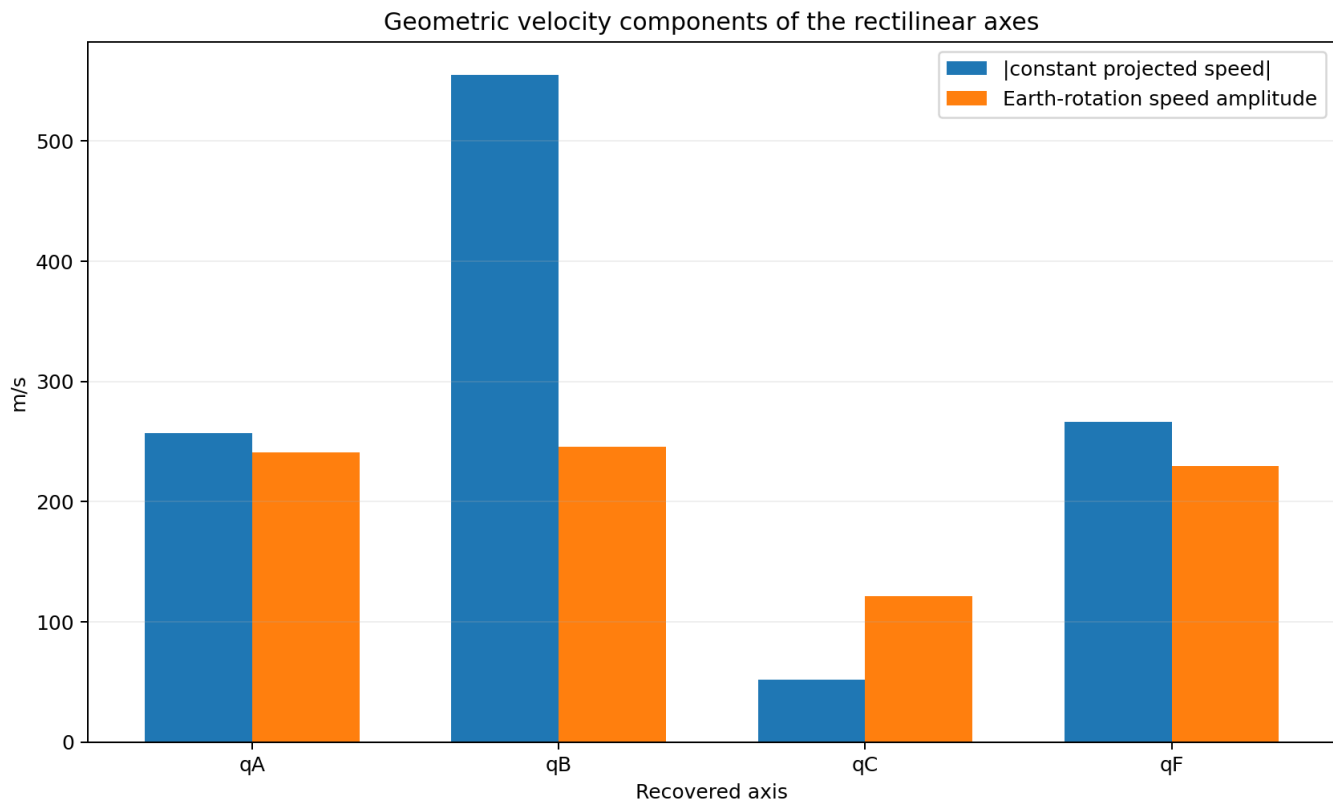


Figure 11. Constant projected speed and Earth-rotation speed amplitude for qA, qB, qC, and qF. The rotational term is comparable to the constant term for qA, qC, and qF, allowing large period variation and velocity reversals.

Axis	Spacing	v0 m/s	Vrot m/s	v min	v max	a amplitude	jerk amplitude
qA	317.864 km	-256.963	240.867	-497.847	-16.095	0.017564	1.280809e-06
qB	1034.800 km	-554.918	245.840	-800.818	-309.070	0.017927	1.307252e-06
qC	260.277 km	-51.529	121.519	-173.059	69.992	0.008861	6.461771e-07
qF	670.886 km	-266.133	229.671	-495.815	-36.460	0.016748	1.221276e-06

8.2 Spatial phase traversal

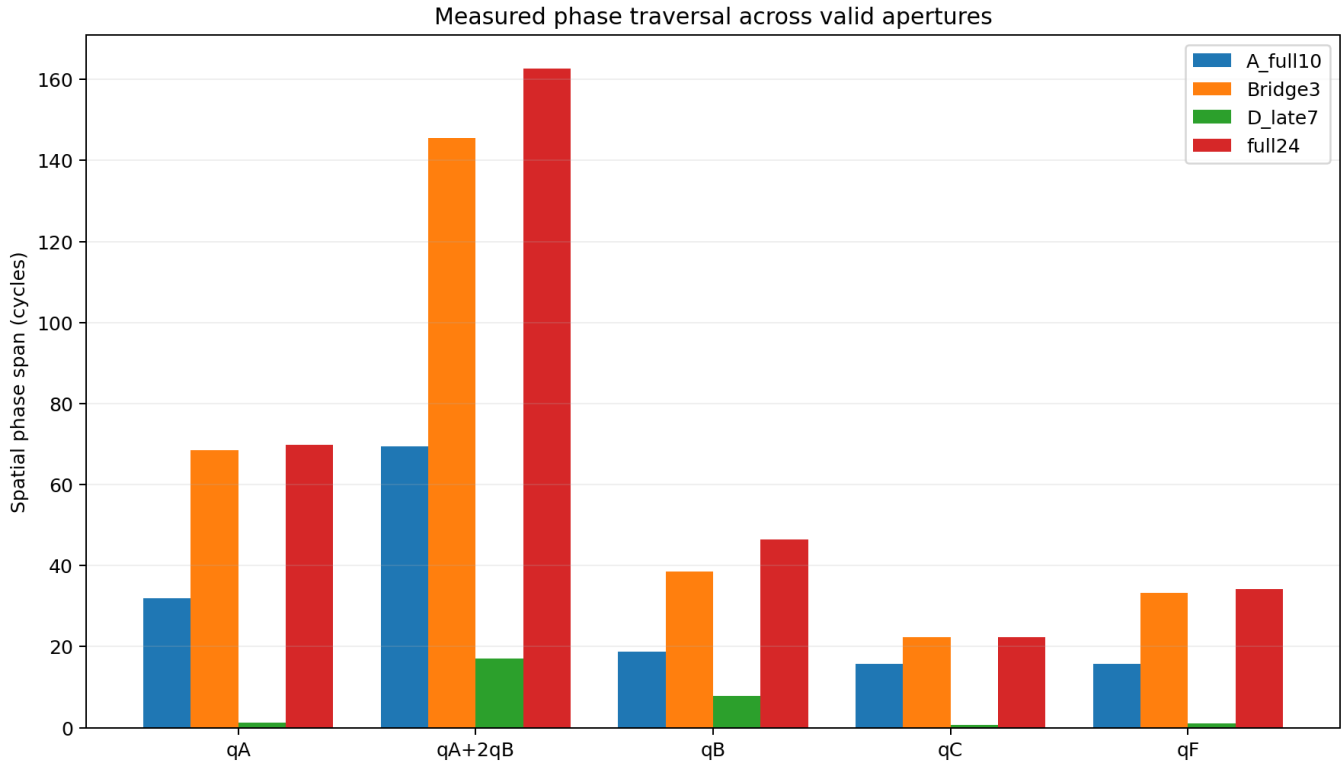


Figure 12. Spatial phase span of each selected axis across the valid apertures. Integer plane crossings and half-sheet crossings are direct counts in the unwrapped coordinate, not counts of repeated local device angles.

Across the full 24-hour trajectory, qA advances approximately 69.75 cycles in magnitude. Different axes traverse different phase spans because their small CMB-parallel components and their transverse Earth-rotation projections differ. The large along-track displacement makes absolute phase exceptionally sensitive to very small parallel-vector errors.

8.3 qC projected-velocity reversals

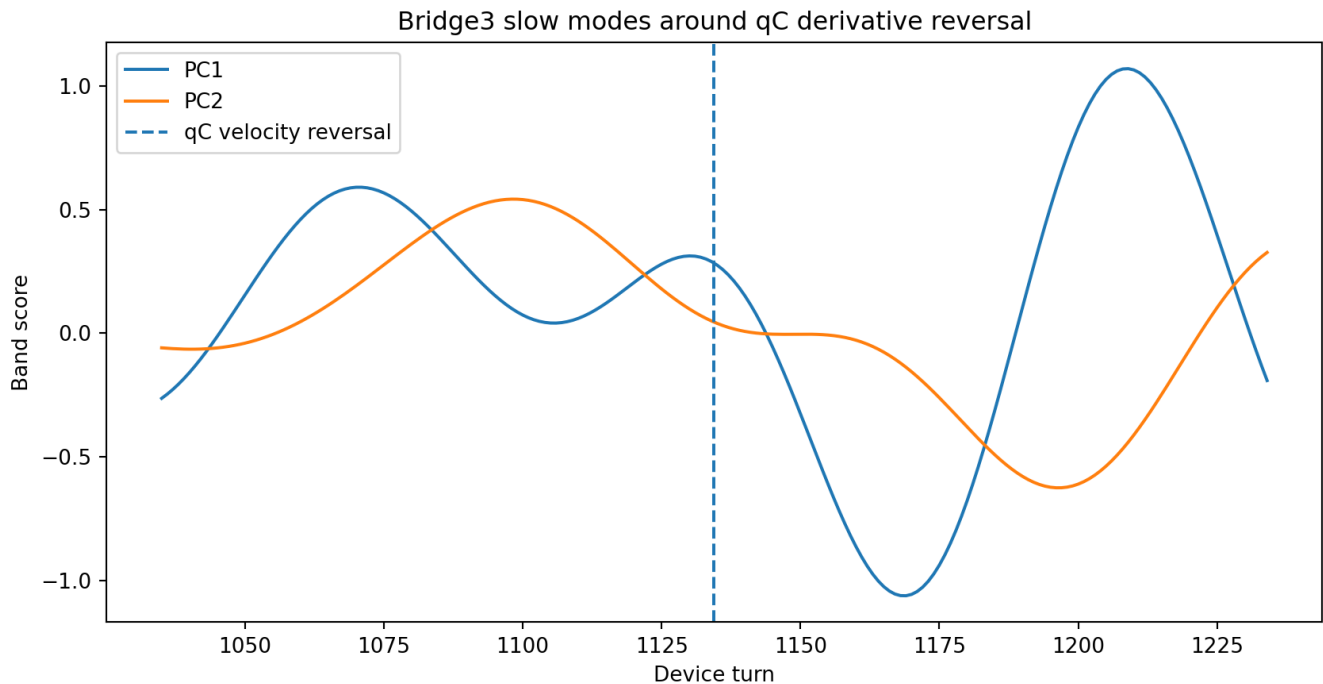


Figure 13. Bridge slow mode near the first qC projected-velocity reversal. The mode derivative changes sign within approximately 4.45 turns of the geometric derivative zero.

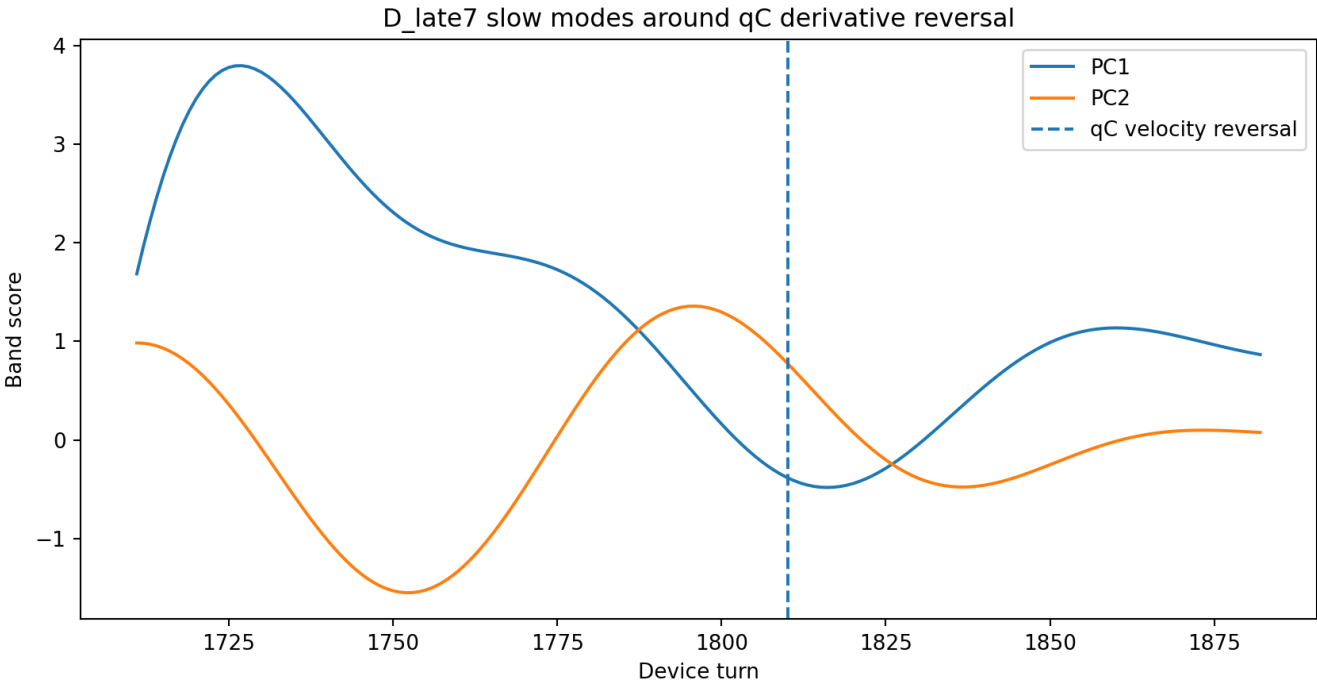


Figure 14. Late seven-channel slow modes near the second qC reversal. Several mode derivatives form nearby extrema or sign changes in a different valid channel regime.

Aperture	Band	PC	qC reversal turn	Nearest mode derivative zero	Offset turns	Mode value	Second derivative
Bridge3	64_300	1	1134.452	1130.000	-4.452	0.28325	-0.003483
Bridge3	64_300	2	1134.452	1147.000	+12.548	0.04536	0.001048
Bridge3	300_1000	1	1134.452	1218.000	+83.548	0.01770	-0.000029
Bridge3	300_1000	2	1134.452	1029.000	-105.452	-0.02504	-0.000018
D_late7	12_64	1	1810.152	1808.000	-2.152	-0.07288	0.031203
D_late7	12_64	2	1810.152	1808.000	-2.152	0.91662	-0.054077
D_late7	64_180	1	1810.152	1816.000	+5.848	-0.38705	0.005108
D_late7	64_180	2	1810.152	1795.000	-15.152	0.76840	-0.001767

9. Full-Signal Rectilinear Path Rankings

The full-valid search compares named reciprocal trajectories across several observables. A path score measures whether time-frequency power is concentrated along the period curve predicted by a fixed reciprocal vector. Because every valid sample is used, the score describes the complete available aperture rather than prospective prediction.

Different signal layers prefer different named trajectories. This is evidence for a multiplex family of rectilinear branches rather than one scalar carrier that dominates every channel combination.

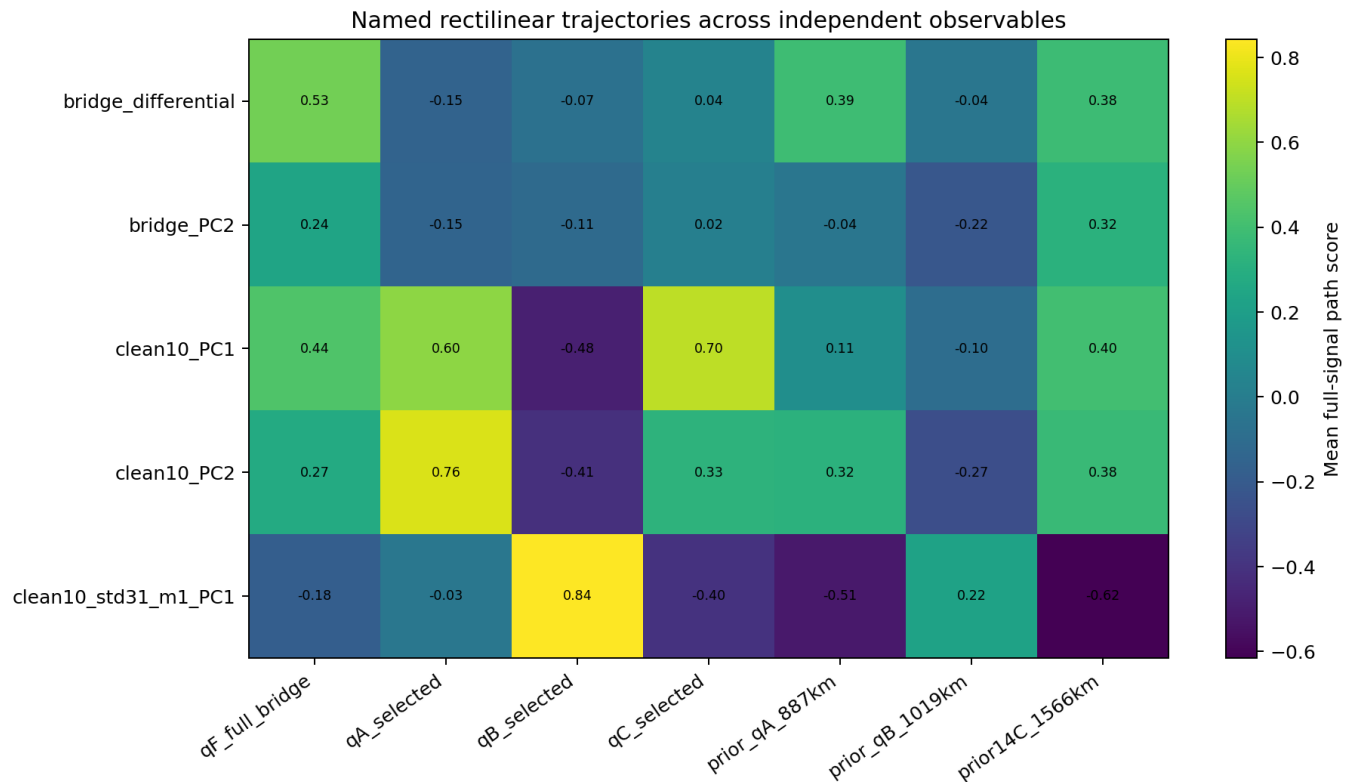


Figure 15. Mean path score for named reciprocal trajectories across independent observables. The bridge differential is strongest on qF; bridge PC2 retains the prior 1,566 km branch; clean spectral PC2 is strongest on qA; clean broadband PC1 is strongest on qC; and the 31-position angular-variance PC1 is strongest on qB.

Observable	Candidate	Mean score	Median score	Positive fraction	Minimum period	Maximum period
bridge_PC2	prior14C_1566km	0.3161	0.4741	0.730	29.80	53.29
bridge_PC2	qF_full_bridge	0.2371	0.3743	0.644	29.43	271.96
bridge_PC2	qC_selected	0.0155	0.0014	0.466	32.71	10555.12
bridge_PC2	prior_qA_887km	-0.0435	0.2303	0.601	30.58	162.36
bridge_PC2	qB_selected	-0.1104	0.0922	0.544	29.47	72.81
bridge_PC2	qA_selected	-0.1483	0.1654	0.310	13.88	186.95
bridge_PC2	prior_qB_1019km	-0.2214	-0.0924	0.452	33.50	123.36
bridge_common	qF_full_bridge	0.2973	0.3158	0.651	29.43	271.96
bridge_common	prior_qA_887km	0.1446	0.2309	0.612	30.58	162.36
bridge_common	prior14C_1566km	0.1126	0.1732	0.630	29.80	53.29
bridge_common	qC_selected	-0.0039	0.0696	0.509	32.71	10555.12
bridge_common	qB_selected	-0.1359	-0.0934	0.470	29.47	72.81
bridge_common	qA_selected	-0.2796	-0.0385	0.274	13.88	186.95
bridge_common	prior_qB_1019km	-0.3739	-0.2139	0.370	33.50	123.36
bridge_differential	qF_full_bridge	0.5343	0.4661	0.811	29.43	271.96
bridge_differential	prior_qA_887km	0.3926	0.4168	0.747	30.58	162.36
bridge_differential	prior14C_1566km	0.3836	0.4205	0.726	29.80	53.29
bridge_differential	qC_selected	0.0387	0.0216	0.473	32.71	10555.12

Observable	Candidate	Mean score	Median score	Positive fraction	Minimum period	Maximum period
bridge_differential	prior_qB_1019km	-0.0447	-0.0121	0.484	33.50	123.36
bridge_differential	qB_selected	-0.0687	-0.0716	0.470	29.47	72.81
bridge_differential	qA_selected	-0.1498	0.0062	0.288	13.88	186.95
clean10_PC1	qC_selected	0.6958	0.8841	0.754	32.71	89.36
clean10_PC1	qA_selected	0.5958	0.6136	0.905	14.44	49.51
clean10_PC1	qF_full_bridge	0.4428	0.5140	0.659	30.39	91.00
clean10_PC1	prior14C_1566km	0.4024	0.4605	0.690	33.06	54.71
clean10_PC1	prior_qA_887km	0.1053	0.1881	0.603	32.15	92.91
clean10_PC1	prior_qB_1019km	-0.0963	0.3705	0.643	40.51	134.43
clean10_PC1	qB_selected	-0.4847	-0.3265	0.302	28.98	52.89
clean10_PC2	qA_selected	0.7593	0.7381	0.937	14.44	49.51
clean10_PC2	prior14C_1566km	0.3756	0.5348	0.762	33.06	54.71
clean10_PC2	qC_selected	0.3299	0.6487	0.738	32.71	89.36
clean10_PC2	prior_qA_887km	0.3219	0.3047	0.627	32.15	92.91
clean10_PC2	qF_full_bridge	0.2741	0.3928	0.675	30.39	91.00
clean10_PC2	prior_qB_1019km	-0.2652	-0.1152	0.460	40.51	134.43
clean10_PC2	qB_selected	-0.4124	-0.1646	0.413	28.98	52.89
clean10_std31_m1_PC1	qB_selected	0.8428	0.8024	0.778	28.98	52.89
clean10_std31_m1_PC1	prior_qB_1019km	0.2235	0.5713	0.698	40.51	134.43
clean10_std31_m1_PC1	qA_selected	-0.0349	0.1334	0.587	14.44	49.51
clean10_std31_m1_PC1	qF_full_bridge	-0.1819	0.0448	0.524	30.39	91.00
clean10_std31_m1_PC1	qC_selected	-0.4021	0.0712	0.579	32.71	89.36
clean10_std31_m1_PC1	prior_qA_887km	-0.5141	0.1171	0.524	32.15	92.91
clean10_std31_m1_PC1	prior14C_1566km	-0.6154	-0.3545	0.310	33.06	54.71
clean10_std31_m1_PC2	qC_selected	0.4475	0.3755	0.746	32.71	89.36
clean10_std31_m1_PC2	prior_qB_1019km	0.4074	0.2764	0.651	40.51	134.43
clean10_std31_m1_PC2	qF_full_bridge	0.2378	0.1512	0.603	30.39	91.00
clean10_std31_m1_PC2	prior14C_1566km	0.0864	0.3142	0.690	33.06	54.71
clean10_std31_m1_PC2	prior_qA_887km	0.0431	0.0981	0.556	32.15	92.91
clean10_std31_m1_PC2	qB_selected	-0.1033	0.0256	0.516	28.98	52.89
clean10_std31_m1_PC2	qA_selected	-0.9048	-0.8471	0.151	14.44	49.51

9.1 Relationship to the earlier 887 km and 1,566 km paths

On the full 445-Clear-NIR differential surface, qF has mean score 0.534, the earlier 887 km path has 0.393, and the earlier 1,566 km path has 0.384. On bridge PC2, the 1,566 km path is the highest-scoring named candidate. The earlier direction therefore remains present, but it occupies a different spectral-state coordinate than the strongest bridge differential.

The correct geometric interpretation is not that one spacing has replaced another. The record contains several populated reciprocal magnitudes on recurring transverse directions. Their relative visibility depends on whether the observable emphasizes common intensity, spectral polarization, local angular variance, or a differential bridge state.

10. Two-Axis Rectilinear Modulation and Torus Embedding

A pair of independent reciprocal phases provides a two-dimensional coordinate system on a torus: each phase is periodic modulo one cycle. If the signal is organized by both plane families, its state should form a structured surface in $(\text{Phi_A mod } 1, \text{Phi_B mod } 1)$ rather than a single sinusoid in either phase alone.

Low-order Fourier bases in these two phases quantify direct harmonics and mixed sum/difference terms. Binned phase maps provide a complementary nonparametric view.

10.1 Low-order reciprocal lattice basis

$$\begin{aligned}\text{Phi_A}(t) &= q_A \cdot R(t) \\ \text{Phi_B}(t) &= q_B \cdot R(t) \\ \text{Basis terms: } &\exp[i 2 \pi (m \text{Phi_A} + n \text{Phi_B})]\end{aligned}$$

For the initial ten-channel 12-64-turn state, q_A harmonics represent 15.45 percent of the multichannel variance, q_B harmonics 4.77 percent, their additive harmonic basis 19.89 percent, and the full low-order two-axis lattice 26.19 percent. For 64-300 turns, direct fundamentals are individually weak while mixed sum/difference terms represent 14.51 percent. The slow state is therefore concentrated partly in intermodulation coordinates.

Aperture	Band	Model	Basis terms	Full-signal variance represented
A_full10	3_12	A	6	1.18%
A_full10	3_12	B	6	0.31%
A_full10	12_64	A	6	15.45%
A_full10	12_64	B	6	4.77%
A_full10	64_300	A	6	0.06%
A_full10	64_300	B	6	0.32%
Bridge3	3_12	A	6	0.41%
Bridge3	3_12	B	6	0.08%
Bridge3	12_64	A	6	4.57%
Bridge3	12_64	B	6	2.19%
Bridge3	64_300	A	6	0.05%
Bridge3	64_300	B	6	0.43%
Bridge3	300_1000	A	6	0.02%
Bridge3	300_1000	B	6	0.05%
D_late7	3_12	A	6	0.02%
D_late7	3_12	B	6	5.57%
D_late7	12_64	A	6	14.71%
D_late7	12_64	B	6	28.49%
D_late7	64_180	A	6	96.59%
D_late7	64_180	B	6	1.25%

Clean ten-channel two-axis rectilinear embedding

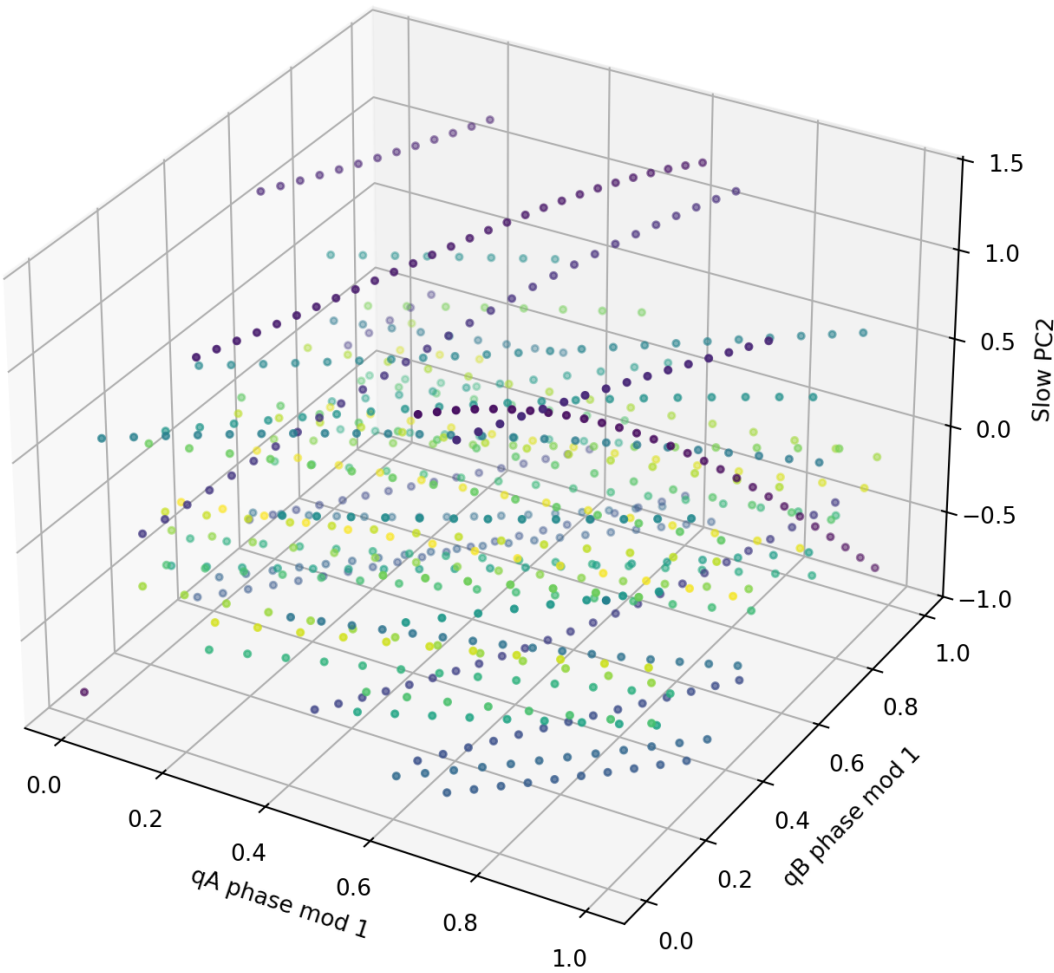


Figure 16. Three-dimensional state embedding using the two rectilinear phases and a spectral state coordinate. The measured trajectory forms a folded surface rather than one closed sinusoidal tube, demonstrating that two spatial phases and a changing multichannel amplitude state are required.

10.2 Two-dimensional phase maps

Aperture	Band	PC	Bins per axis	Occupied bins	Phase-map R2
A_full10	64_300	1	12	128	0.361
A_full10	64_300	2	12	128	0.432
Bridge3	300_1000	1	18	297	0.401
Bridge3	300_1000	2	18	297	0.321
D_late7	64_180	1	10	51	0.978
D_late7	64_180	2	10	51	0.991

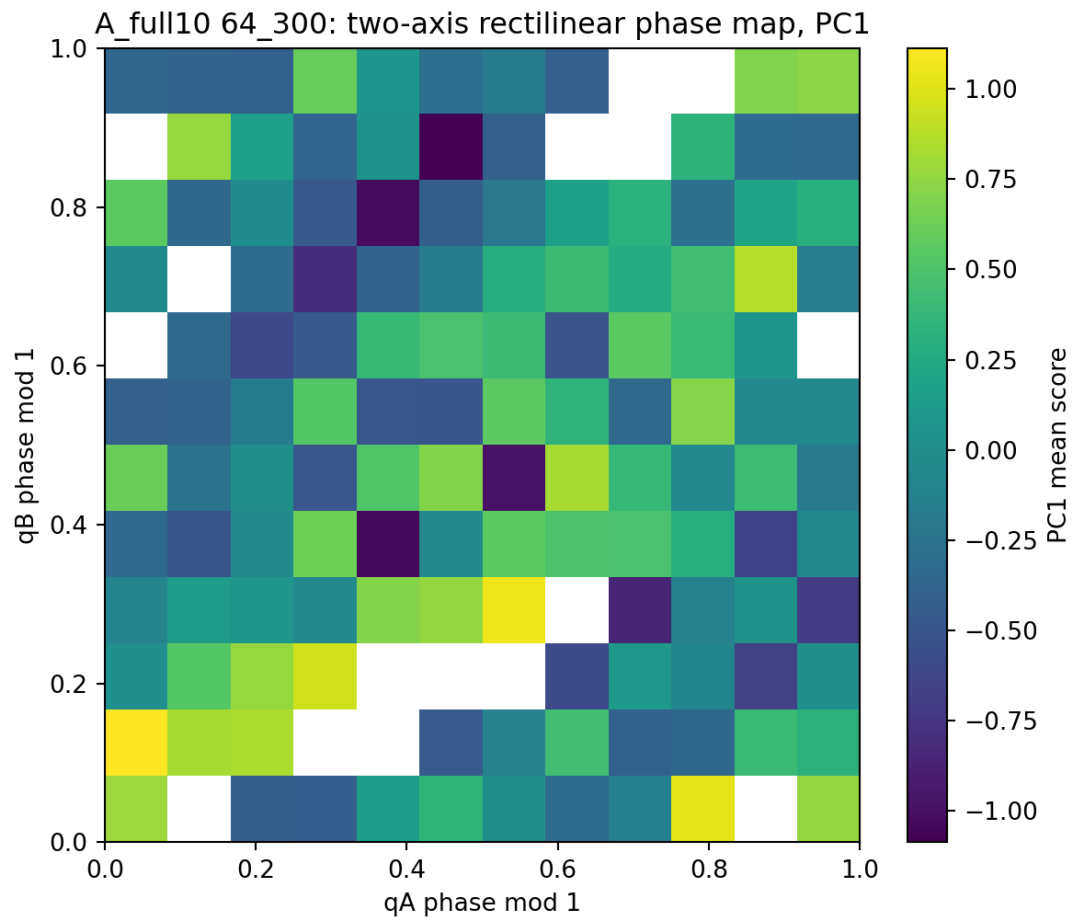


Figure 17. Initial ten-channel slow PC1 mean state on the (Φ_A , Φ_B) phase torus. Occupied phase bins form a structured surface and represent approximately 36.1 percent of the full signal.

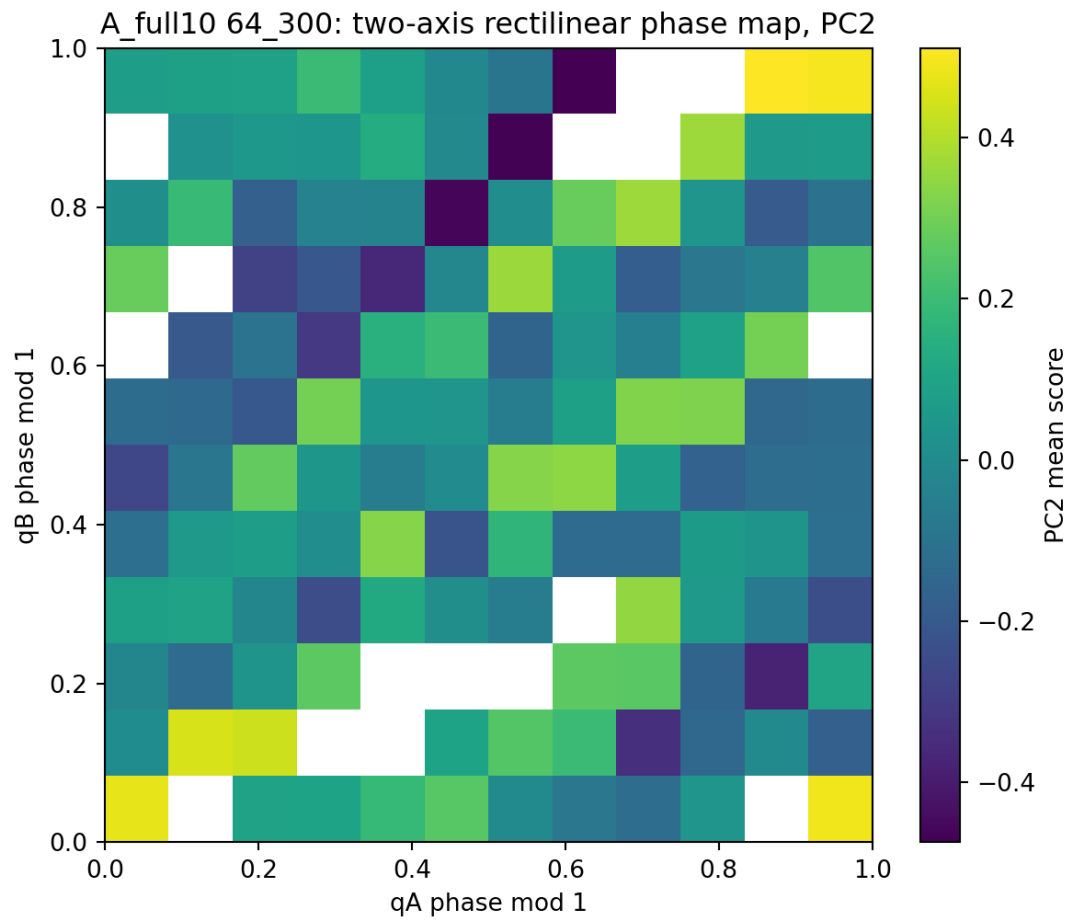


Figure 18. Initial ten-channel slow PC2 on the same torus. The spectral-polarization state has a distinct two-axis phase texture and approximately 43.2 percent phase-map representation.

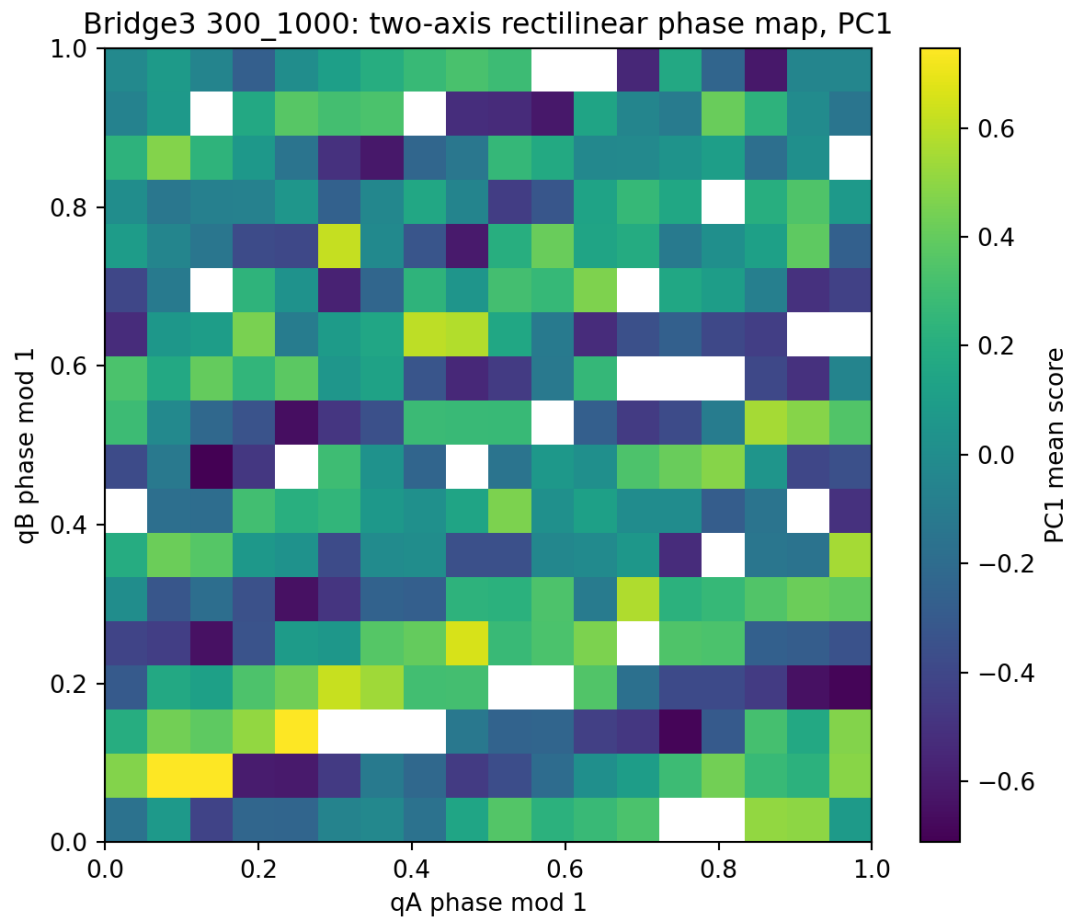


Figure 19. Ultraslow bridge PC1 on the two-axis phase torus. The long three-channel state retains a structured dependence on both phases.

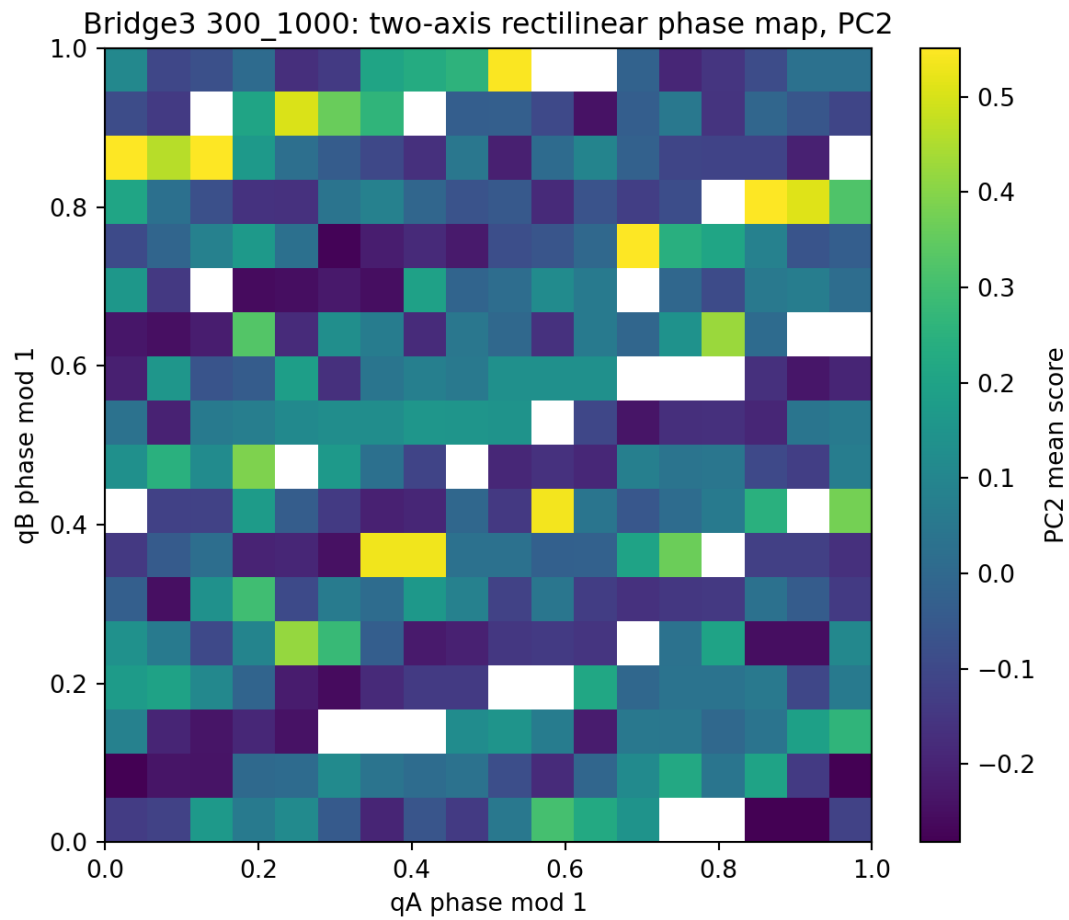


Figure 20. Ultraslow bridge PC2 on the two-axis phase torus. This map complements the complex phase rotation described later and shows that the bridge state is not reducible to one rectilinear coordinate.

11. Low-Order Reciprocal-Vector Grammar

The selected qC broadband direction is closely approximated by a low-order combination of the qA spectral-polarization direction and the qB angular-variance direction. This provides a compact algebraic relation among independently extracted modes.

The relation must be read at two levels. Locally, it is a tight vector closure in direction and magnitude. Globally, its small residual is amplified by the enormous CMB translation and prevents exact absolute phase registry.

$$qC \sim qA + 2 qB$$

$$\text{relative vector mismatch} = |qC - (qA + 2 qB)| / |qC| = 0.013029$$

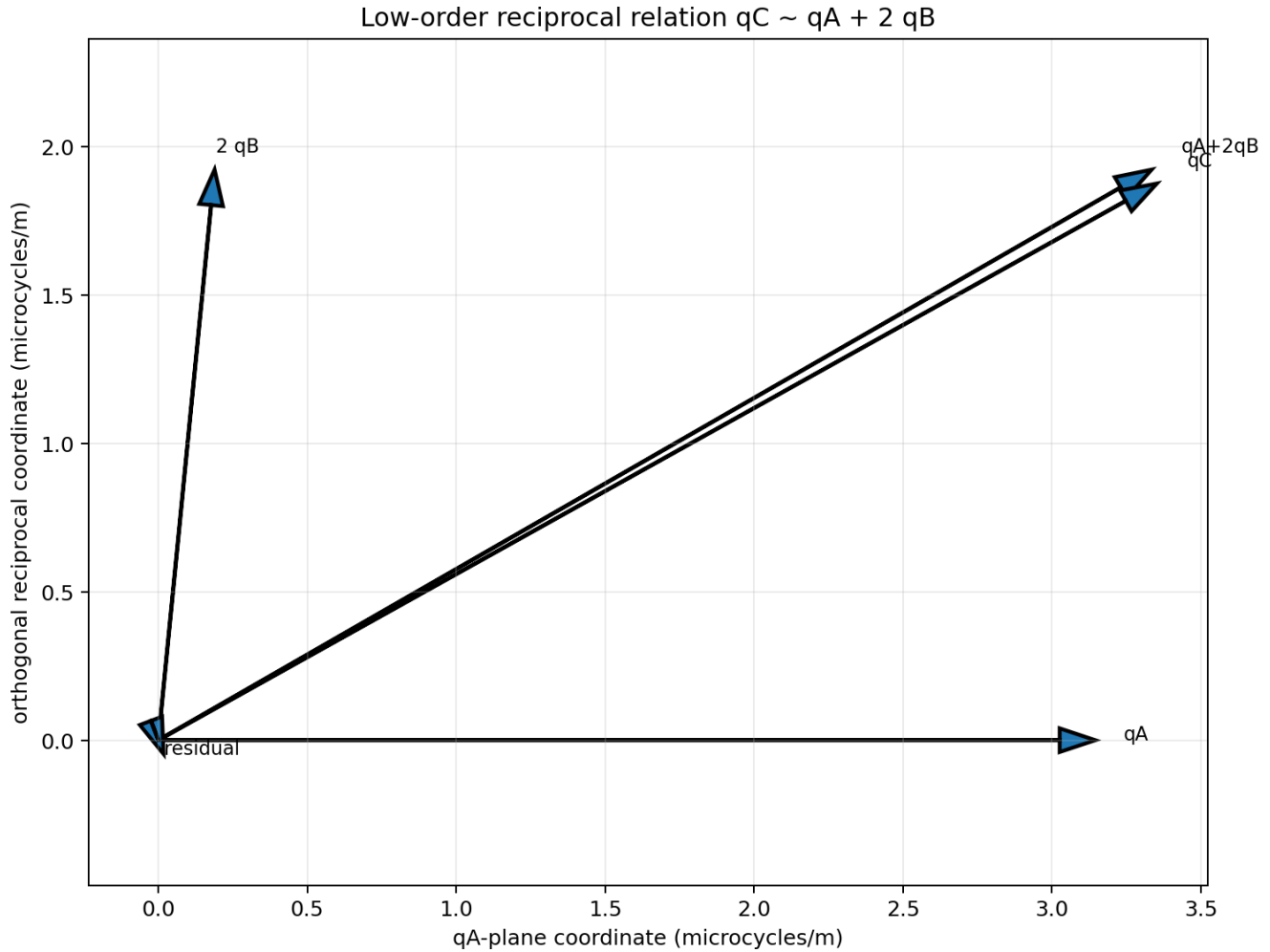


Figure 21. Reciprocal-vector construction in the qA-qB plane. qC lies close to qA + 2 qB. The residual is small compared with qC but not zero.

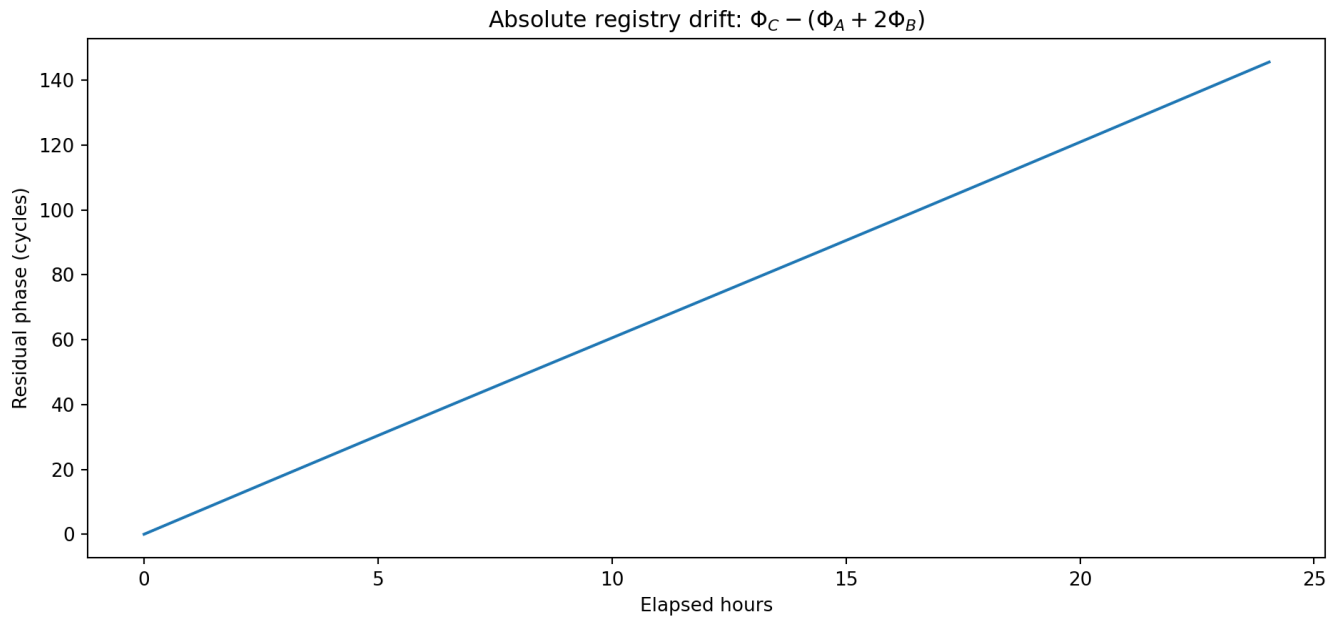


Figure 22. Accumulated phase drift of the $q_C - q_A - 2q_B$ residual. A small reciprocal mismatch becomes a large phase mismatch over the approximately 80-billion-metre translation, separating local vector grammar from global lattice registry.

The residual accumulates approximately 146 cycles across the full record. To preserve one-cycle registry across the complete translation, the CMB-parallel reciprocal error would need to be of order 10^{-11} cycles per metre or smaller. The present direction estimates are not constrained at that level.

Vector error	m	n	A observable	A km	B observable	B km	C observable	C km	Full residual phase cycles
0.283%	2	1	bridge_differential	768.02	bridge_PC2	1860.85	clean10_PC2	317.86	14.64
0.283%	-2	-1	bridge_differential	768.02	bridge_PC2	1860.85	clean10_PC2	317.86	14.64
1.010%	2	-2	bridge_common	364.14	clean10_PC1	260.28	clean10_PC2	345.58	140.60
1.010%	-2	2	bridge_common	364.14	clean10_PC1	260.28	clean10_PC2	345.58	140.60
1.063%	-3	-1	bridge_common	1659.06	bridge_PC2	508.23	clean10_PC2	345.58	106.67
1.063%	3	1	bridge_common	1659.06	bridge_PC2	508.23	clean10_PC2	345.58	106.67
1.364%	4	-3	clean10_PC2	620.65	clean10_std31_m1_PC2	390.95	bridge_PC2	810.15	524.17
1.364%	-4	3	clean10_PC2	620.65	clean10_std31_m1_PC2	390.95	bridge_PC2	810.15	524.17
1.455%	-1	-2	bridge_common	389.10	bridge_PC2	508.23	clean10_std31_m1_PC1	156.74	32.54
1.455%	1	2	bridge_common	389.10	bridge_PC2	508.23	clean10_std31_m1_PC1	156.74	32.54
1.563%	-3	1	bridge_common	1659.06	clean10_PC2	345.58	bridge_PC2	508.23	106.67
1.563%	3	-1	bridge_common	1659.06	clean10_PC2	345.58	bridge_PC2	508.23	106.67
1.579%	-4	-1	bridge_differential	1251.26	bridge_PC2	810.15	clean10_PC1	260.28	111.53
1.579%	4	1	bridge_differential	1251.26	bridge_PC2	810.15	clean10_PC1	260.28	111.53
1.659%	-2	1	bridge_differential	768.02	clean10_PC2	317.86	bridge_PC2	1860.85	14.64
1.659%	2	-1	bridge_differential	768.02	clean10_PC2	317.86	bridge_PC2	1860.85	14.64
1.787%	3	0	bridge_common	389.10	clean10_std31_m1_PC2	390.95	clean10_PC2	128.01	185.94
1.787%	-3	0	bridge_common	389.10	clean10_std31_m1_PC1	156.74	clean10_PC2	128.01	185.94
1.787%	-3	0	bridge_common	389.10	bridge_PC2	1860.85	clean10_PC2	128.01	185.94
1.787%	0	-3	bridge_differential	768.02	bridge_common	389.10	clean10_PC2	128.01	185.94

12. Complex Spectral Phase and Hidden Channel Organization

Channel information is not confined to intensity. The phase graph often contains a clearer ordering than the amplitude graph. At longer periods, channels arrange along a smooth progression with photon energy and form antipodal and quadrature relationships that can be nearly invisible to ordinary linear correlation.

The synchronized phase of each channel is obtained from the leading complex phase-graph eigenvector. A low phase-graph participation dimension means that many pairwise phase relations can be explained by a small number of channel phase coordinates.

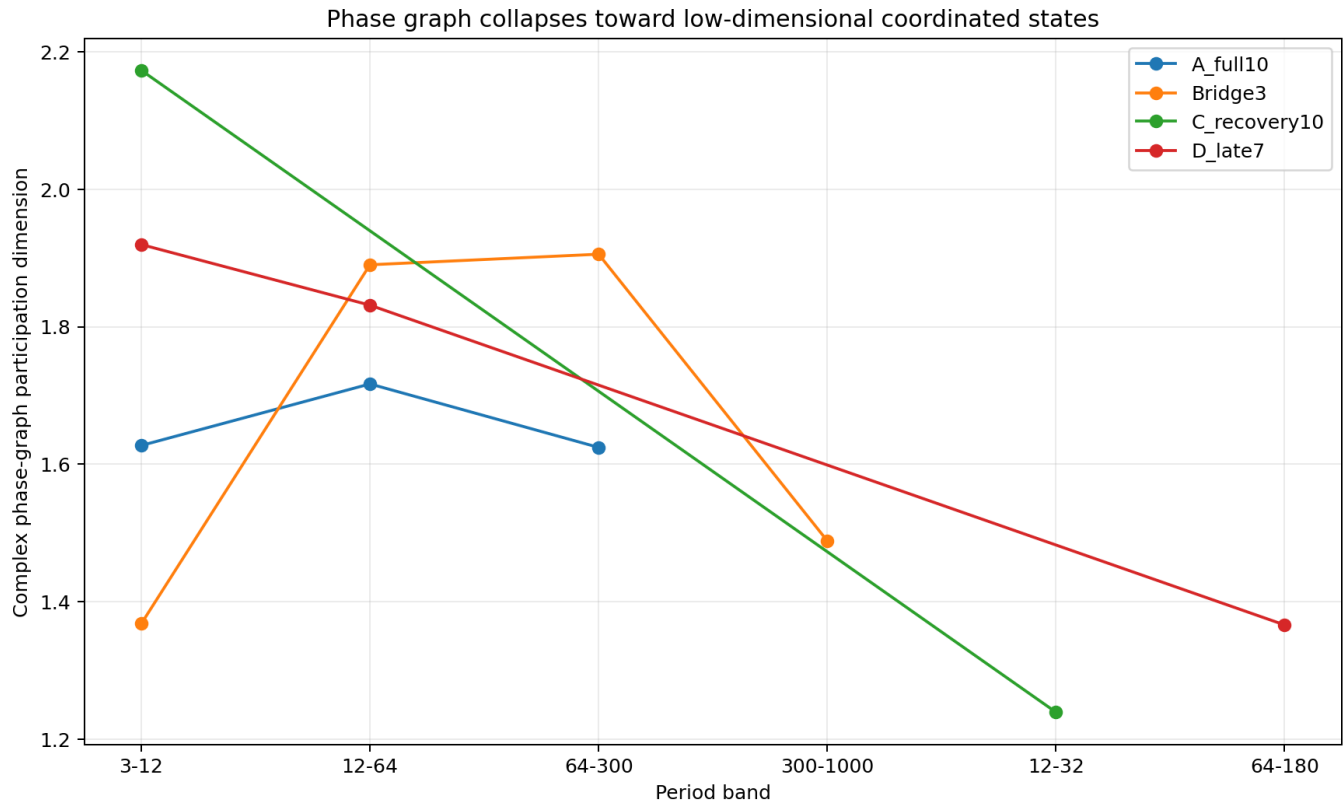


Figure 23. Complex phase-graph participation dimension across valid apertures and period bands. Slow states collapse toward a coordinated low-dimensional phase organization even when amplitude participation dimension increases.

Aperture	Band	Lead eig.	Second eig.	Graph dimension	Phase residual	Triangle holonomy	Phase-energy R2
A_full10	12_64	0.746	0.156	1.717	8.78 deg	18.29 deg	0.543
A_full10	3_12	0.779	0.060	1.627	1.21 deg	2.35 deg	0.060
A_full10	64_300	0.763	0.179	1.624	10.03 deg	20.25 deg	0.709
Bridge3	12_64	0.665	0.290	1.890	13.91 deg	46.21 deg	nan
Bridge3	300_1000	0.797	0.192	1.488	9.59 deg	29.25 deg	nan
Bridge3	3_12	0.848	0.097	1.368	0.19 deg	0.57 deg	nan
Bridge3	64_300	0.656	0.304	1.905	12.31 deg	44.16 deg	nan
C_recovery10	12_32	0.893	0.098	1.239	3.07 deg	5.97 deg	0.000
C_recovery10	3_12	0.652	0.152	2.173	7.53 deg	15.01 deg	0.400
D_late7	12_64	0.720	0.144	1.831	4.56 deg	10.46 deg	0.438
D_late7	3_12	0.704	0.142	1.920	3.84 deg	10.08 deg	0.614
D_late7	64_180	0.841	0.156	1.366	6.74 deg	17.55 deg	0.393

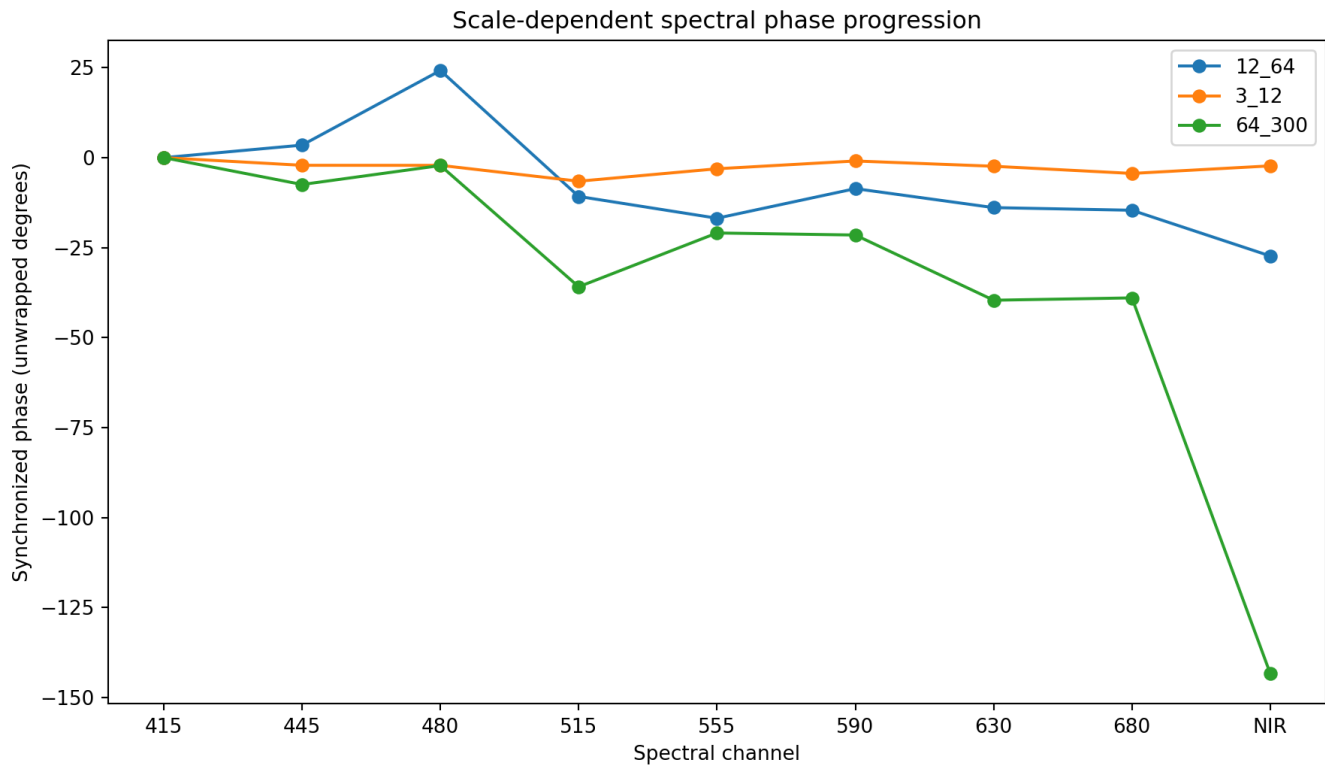


Figure 24. Synchronized channel phase versus photon energy. The relation strengthens with period: the slow band develops an ordered, curved phase progression that a cubic energy-phase representation captures with R2 approximately 0.959.

12.1 Antipodal and quadrature edges

Strong slow antipodal relationships include 480-NIR with coherence 0.758 at 170.07 degrees, 415-NIR with coherence 0.726 at 162.92 degrees, and 445-NIR with coherence 0.578 at 160.36 degrees. A particularly hidden quadrature relation is 555-NIR: Pearson correlation is approximately 0.0034, complex coherence is 0.424, and phase separation is 90.11 degrees.

Aperture	Band	Pair	Coherence	Phase	Class
A_full10	64_300	480-NIR	0.7580	+170.07 deg	antipodal
A_full10	64_300	415-NIR	0.7263	+162.92 deg	antipodal
A_full10	64_300	445-NIR	0.5783	+160.36 deg	antipodal
A_full10	64_300	555-NIR	0.4242	+90.11 deg	quadrature
Bridge3	300_1000	clear-NIR	0.5296	+70.02 deg	quadrature
D_late7	64_180	515-630	0.4653	-108.49 deg	quadrature

12.2 Continuous phase rotation across scale

The 445-NIR bridge pair rotates from near-common phase at short periods through intermediate quadrature toward a broad ultraslow antipodal state. This is a scale-dependent rotation of the latent spectral coordinate, not merely a change in correlation magnitude. Clear remains closer to 445 nm, making the three-channel bridge a useful basis for separating common and polarized states.

13. Hidden Nonlinear Relations and Quadrature Loops

Pearson correlation measures one straight-line projection. A pair of channels separated by approximately 90 degrees can trace an ellipse or loop and have almost zero Pearson correlation even when the pair is strongly coupled. The full-valid analysis therefore ranks pairs by the gain obtained when a quadrature coordinate is added to the linear model.

The strongest relations occur in several apertures and bands, demonstrating that complex modulation is distributed across the channel graph rather than confined to one pair.

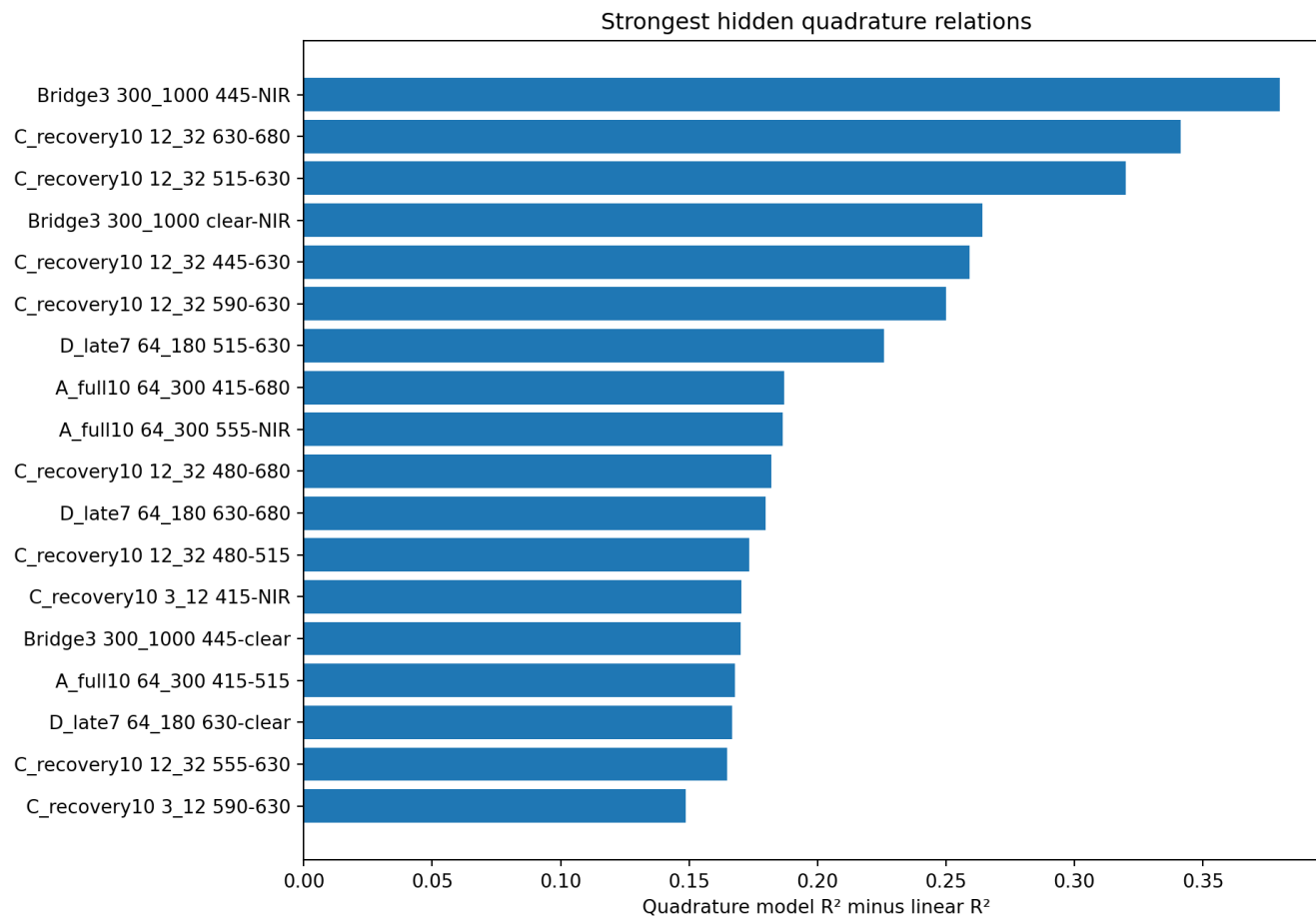


Figure 25. Ranked gain of quadrature models over single-slope linear models. Large gain identifies channel pairs whose information is hidden in phase rotation or curved joint-state trajectories.

Aperture	Band	Pair	Pearson	Mutual info	Complex coh.	Phase deg	Quadrature R2	Gain over linear
Bridge3	300_1000	445-NIR	-0.4268	1.5249	0.7409	+128.15	0.5621	0.3799
C_recovery10	12_32	630-680	+0.5888	0.3551	0.8275	-44.80	0.6881	0.3414
C_recovery10	12_32	515-630	+0.6427	0.3278	0.8545	+41.37	0.7329	0.3199
Bridge3	300_1000	clear-NIR	+0.1899	1.3384	0.5296	+70.02	0.3003	0.2642
C_recovery10	12_32	445-630	+0.5472	0.3283	0.7469	+42.94	0.5585	0.2591
C_recovery10	12_32	590-630	+0.5399	0.3427	0.7352	+42.81	0.5416	0.2501
D_late7	64_180	515-630	-0.2007	0.9735	0.4653	-108.49	0.2662	0.2259
A_full10	64_300	415-680	+0.4686	1.4773	0.6320	+42.53	0.4065	0.1869
A_full10	64_300	555-NIR	+0.0034	1.2192	0.4242	+90.11	0.1865	0.1865
C_recovery10	12_32	480-680	+0.6425	0.4251	0.7687	-33.64	0.5950	0.1821
D_late7	64_180	630-680	+0.8661	1.0089	0.9621	-25.96	0.9299	0.1799
C_recovery10	12_32	480-515	+0.6962	0.3852	0.8091	-30.93	0.6581	0.1735
C_recovery10	3_12	415-NIR	+0.5366	0.1462	0.6766	-37.58	0.4582	0.1703
Bridge3	300_1000	445-clear	+0.7447	2.2661	0.8331	+28.88	0.7245	0.1699

Aperture	Band	Pair	Pearson	Mutual info	Complex coh.	Phase deg	Quadrature R2	Gain over linear
A_full10	64_300	415-515	+0.5717	1.4721	0.6997	+35.31	0.4946	0.1678
D_late7	64_180	630-clear	+0.8368	1.2029	0.9305	-25.63	0.8670	0.1668
C_recovery10	12_32	555-630	+0.6514	0.2956	0.7672	+31.94	0.5891	0.1648
C_recovery10	3_12	590-630	+0.4607	0.1756	0.6009	-39.85	0.3609	0.1486
C_recovery10	3_12	445-630	+0.4645	0.1900	0.6019	-39.49	0.3626	0.1468
A_full10	64_300	415-clear	+0.7575	1.8023	0.8458	+26.31	0.7185	0.1447
D_late7	64_180	630-NIR	+0.8431	1.0918	0.9229	-23.71	0.8529	0.1422
C_recovery10	3_12	515-630	+0.4689	0.1950	0.5989	-38.63	0.3599	0.1400
A_full10	64_300	clear-NIR	-0.1656	1.2692	0.4000	+116.20	0.1617	0.1342
C_recovery10	3_12	555-NIR	+0.7166	0.2988	0.8036	-26.93	0.6460	0.1325
D_late7	64_180	590-630	+0.8746	1.2660	0.9425	+21.62	0.8890	0.1240
A_full10	64_300	445-680	+0.6392	1.3358	0.7259	+28.75	0.5324	0.1238
A_full10	64_300	680-NIR	+0.1775	1.2262	0.3850	+63.61	0.1548	0.1232
C_recovery10	3_12	445-NIR	+0.6989	0.2559	0.7817	-26.64	0.6113	0.1229
A_full10	64_300	480-680	+0.4433	1.4439	0.5589	+38.05	0.3185	0.1220
C_recovery10	12_32	630-NIR	+0.4804	0.2374	0.5919	-35.78	0.3506	0.1198

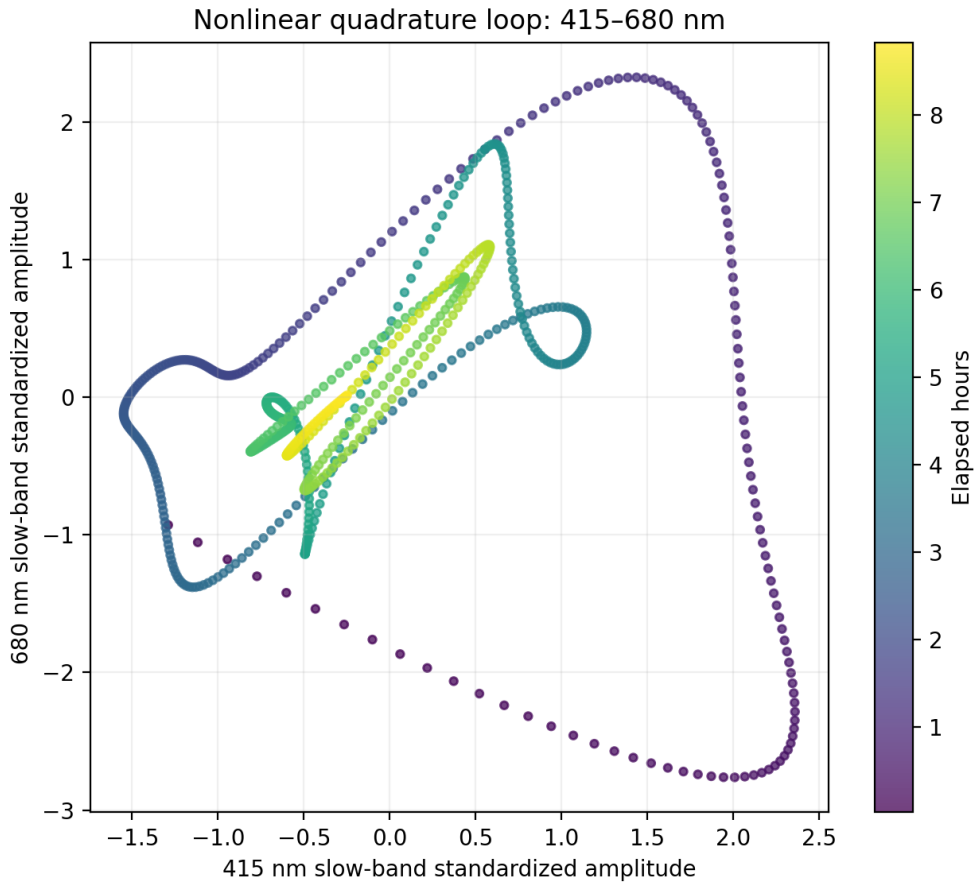


Figure 26. Actual 415-680 nm joint-state loop. The pair has weak linear correlation but strong complex coherence and approximately quadrature phase, so its relationship appears as a curved trajectory rather than a line.

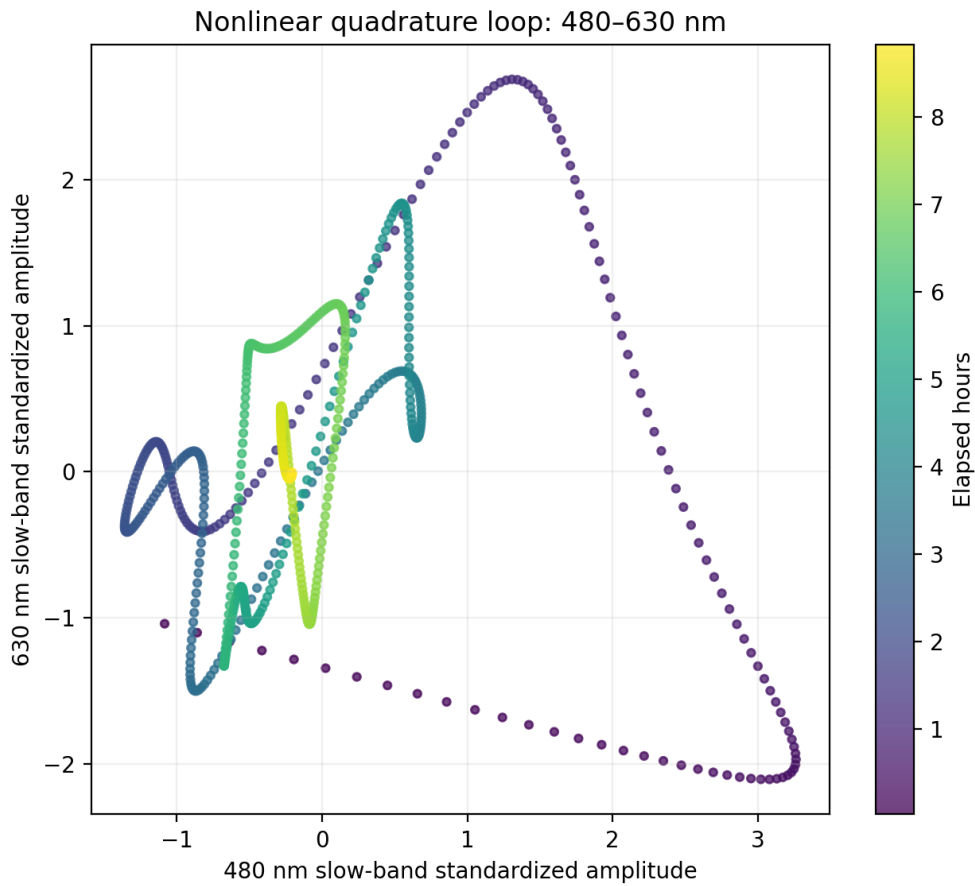


Figure 27. Actual 480-630 nm nonlinear loop. The coupled state is visible in the geometry of the joint trajectory even when a one-dimensional linear projection loses it.

13.1 Interpretation of quadrature coordinates

A quadrature pair can be interpreted as two projections of a rotating latent state. If channel i is approximately $\cos(\phi)$ and channel j is approximately $\sin(\phi)$, their scatter is an ellipse and their zero-lag Pearson coefficient can approach zero. Complex coherence, analytic phase, and two-coordinate regression recover the relation. This is why multiple channels can hide one spatial modulation among phase-shifted responses.

14. Nontrivial Phase Locks and Late-State Frequency Division

Generalized phase locking tests the combination $m\phi_i - n\phi_j$. A strong 2:1 lock means that two cycles of one channel phase remain aligned with one cycle of another, up to a nearly fixed offset. This is distinct from ordinary 1:1 coherence.

During the final seven-channel aperture, 515 nm separates from the red/NIR group and forms strong 2:1 locks with several channels. This is consistent with a subharmonic or frequency-division coordinate centered on the 515 nm state.

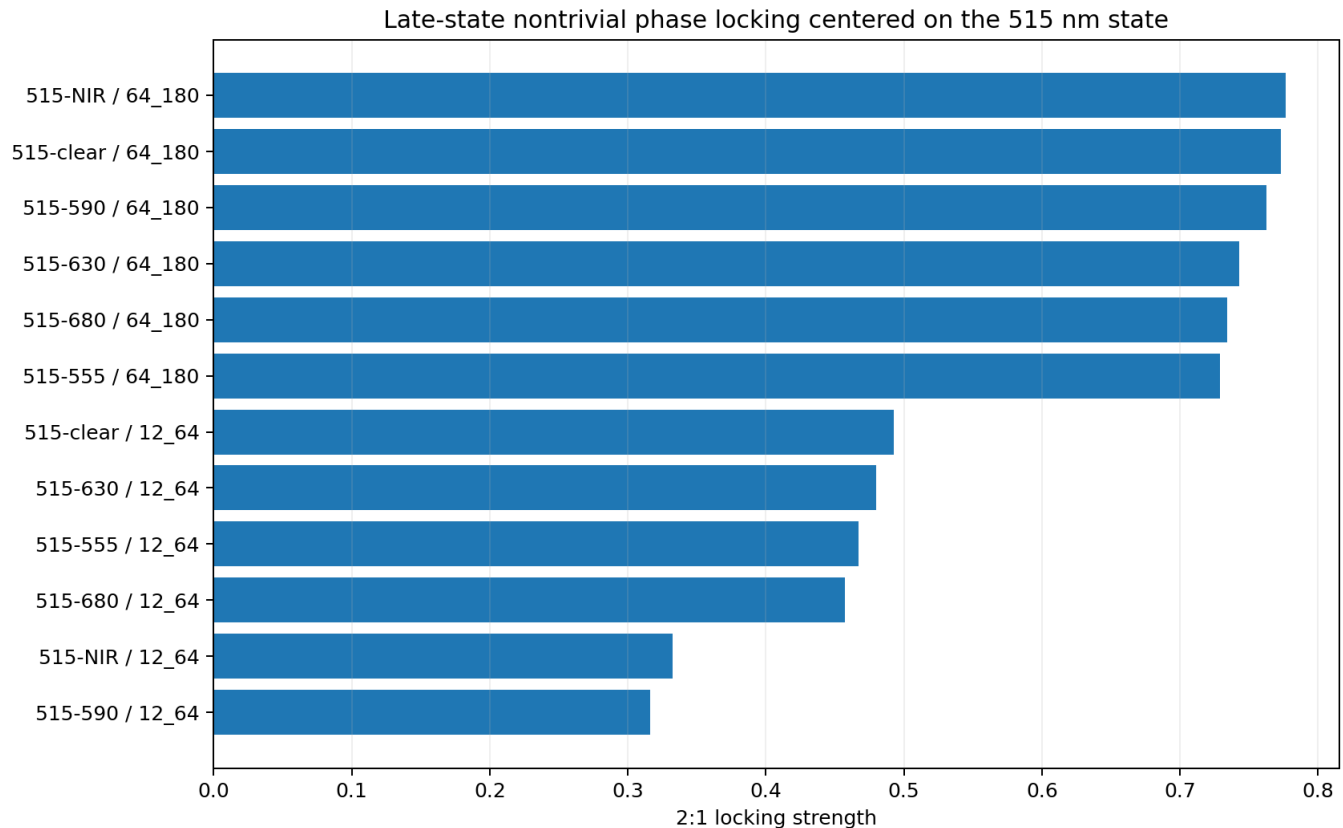


Figure 28. Strongest late-state 2:1 phase locks. The 515 nm channel is repeatedly paired with NIR, Clear, and red-visible channels, indicating a coherent harmonic state rather than independent band noise.

Aperture	Band	Pair	m	n	Lock strength	Offset
D_late7	64_180	515-NIR	2	1	0.7765	-121.91 deg
D_late7	64_180	515-clear	2	1	0.7732	-122.70 deg
D_late7	64_180	515-590	2	1	0.7627	-116.83 deg
D_late7	64_180	515-630	2	1	0.7432	-99.53 deg
D_late7	64_180	515-680	2	1	0.7342	-121.32 deg
D_late7	64_180	515-555	2	1	0.7291	-108.52 deg
Bridge3	300_1000	clear-NIR	1	2	0.6414	+110.87 deg
Bridge3	300_1000	clear-NIR	2	3	0.6045	+155.30 deg
D_late7	64_180	515-555	3	1	0.5796	-113.13 deg
D_late7	64_180	515-680	3	1	0.5658	-125.46 deg
D_late7	64_180	515-590	3	1	0.5613	-122.16 deg
A_full10	64_300	590-NIR	1	2	0.5484	+25.32 deg
D_late7	64_180	515-clear	3	1	0.5459	-127.11 deg
D_late7	64_180	515-NIR	3	1	0.5356	-126.03 deg
Bridge3	300_1000	445-NIR	1	2	0.5320	+146.84 deg
Bridge3	300_1000	445-NIR	1	3	0.5234	-28.84 deg
Bridge3	300_1000	clear-NIR	1	3	0.5129	-26.45 deg
A_full10	64_300	480-590	3	2	0.5114	+2.11 deg

Aperture	Band	Pair	m	n	Lock strength	Offset
A_full10	64_300	555-NIR	1	2	0.5010	+29.97 deg
D_late7	12_64	515-clear	2	1	0.4925	+139.19 deg
D_late7	64_180	515-630	3	1	0.4923	-98.90 deg
C_recovery10	12_32	480-NIR	3	2	0.4882	+121.65 deg
D_late7	12_64	515-630	2	1	0.4798	+123.30 deg
D_late7	12_64	515-555	2	1	0.4672	+132.45 deg
A_full10	64_300	680-NIR	2	3	0.4647	+74.60 deg
Bridge3	300_1000	445-clear	3	2	0.4604	-100.18 deg
Bridge3	300_1000	445-NIR	2	3	0.4577	-136.07 deg
A_full10	64_300	630-NIR	1	2	0.4571	-5.34 deg
D_late7	12_64	515-680	2	1	0.4570	+114.14 deg
C_recovery10	12_32	480-590	3	2	0.4515	+103.86 deg
A_full10	64_300	480-NIR	3	1	0.4482	-179.86 deg
A_full10	64_300	480-630	3	2	0.4410	+39.11 deg
A_full10	64_300	415-590	3	2	0.4379	-14.48 deg
A_full10	64_300	480-680	1	2	0.4329	+74.38 deg
C_recovery10	12_32	445-480	2	3	0.4315	-94.25 deg
Bridge3	300_1000	445-clear	2	1	0.4259	-137.23 deg
A_full10	64_300	680-NIR	1	2	0.4237	+11.05 deg
A_full10	64_300	415-clear	2	3	0.4214	+96.16 deg
A_full10	64_300	515-NIR	2	3	0.4180	+87.50 deg
A_full10	64_300	clear-NIR	3	2	0.4176	-65.88 deg

At 64-180 turns, the synchronized 515 nm phase lies near zero while 630, 555, 590, 680, NIR, and Clear occupy approximately 123 to 151 degrees. The 2:1 combination folds this separation into a common harmonic relation. This is one example of information that is hidden if channels are compared only at equal frequency and equal phase order.

15. Sidebands, Carrier Envelopes, and Phase-Amplitude Coupling

The bridge spectrum contains many peaks whose frequencies satisfy low-order sum relations. Such relations are expected when a carrier is amplitude- or phase-modulated by a slower envelope. They also arise from mixed rectilinear phases $m \Phi_A + n \Phi_B$.

The full-signal exploratory analysis reports the arithmetic structure directly. It does not require that every triad be assigned to one physical nonlinearity.

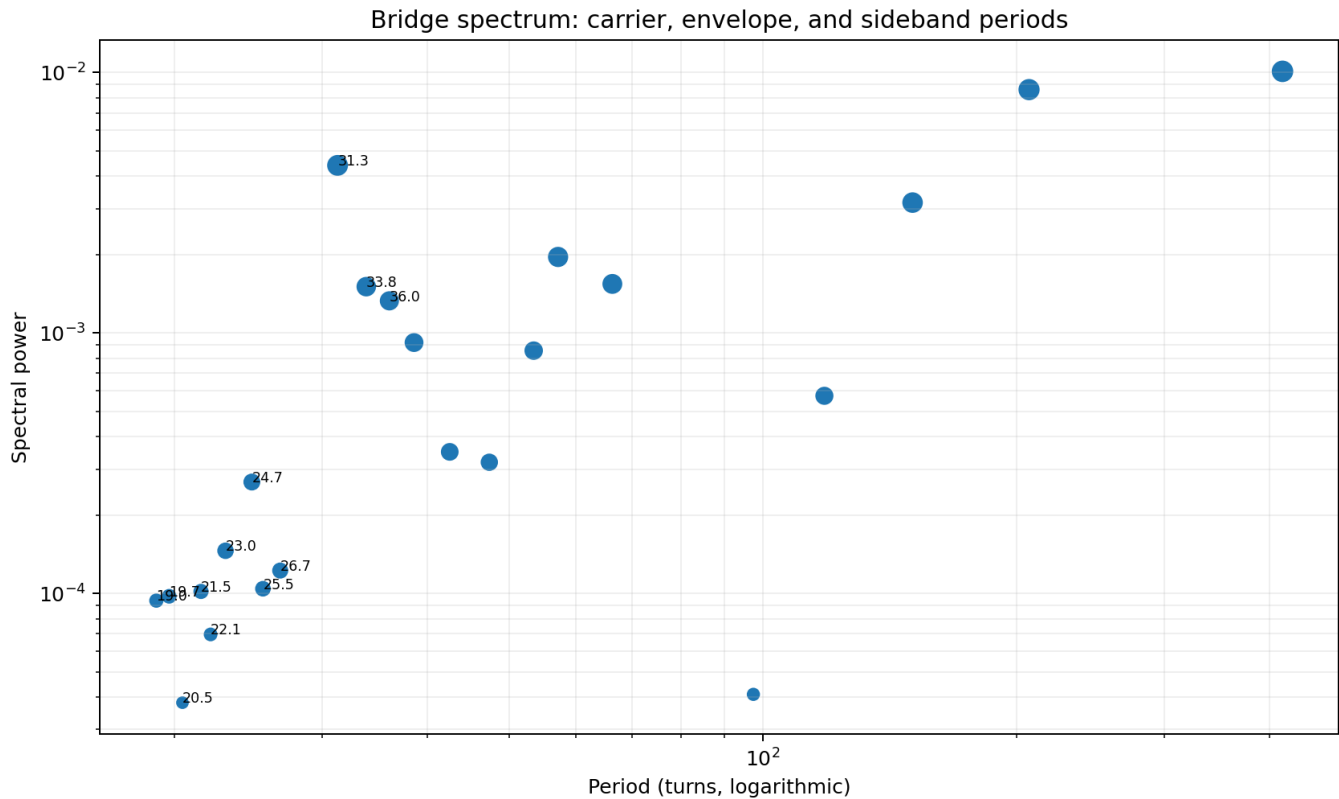


Figure 29. Bridge spectral peaks across the full valid 1,657-turn aperture. Periods are shown on logarithmic axes to expose the hierarchy from tens to hundreds of turns.

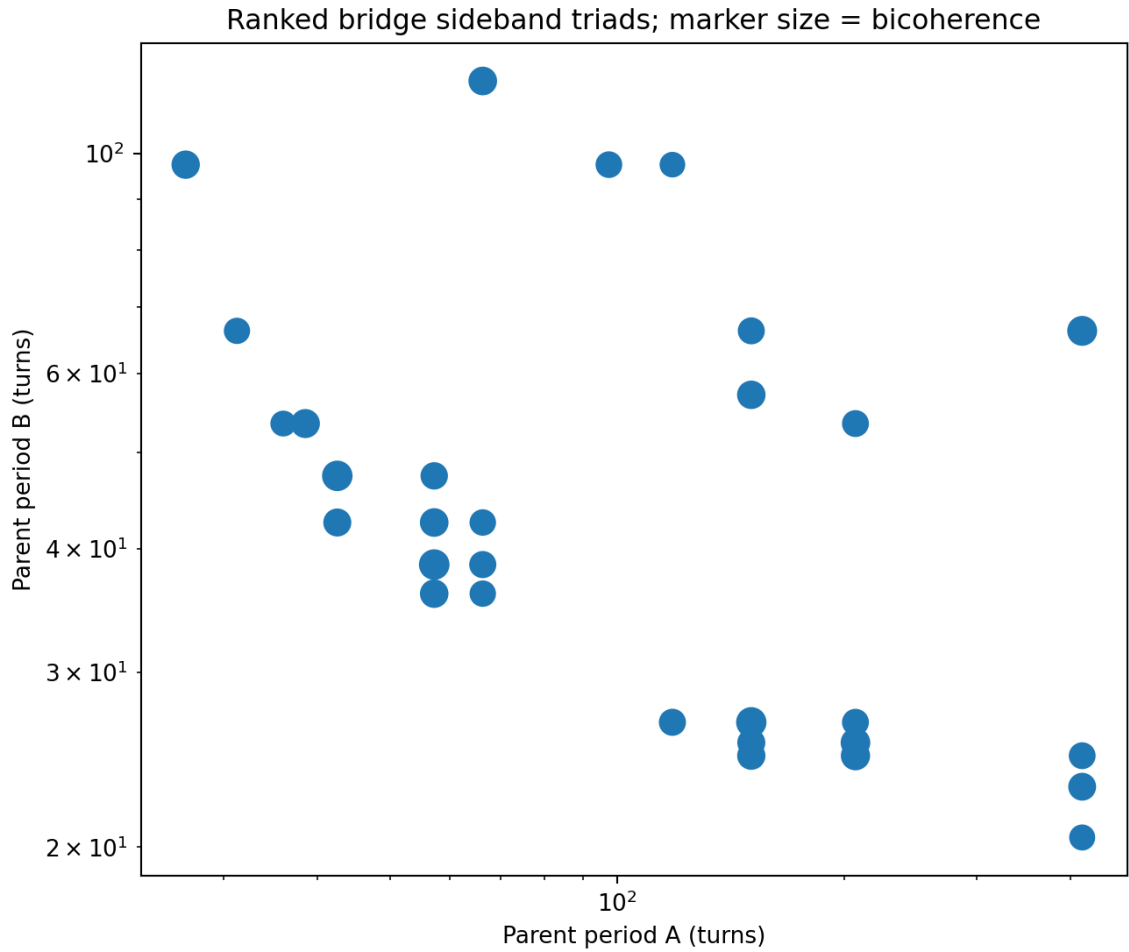


Figure 30. High-ranked sum-frequency triads in the bridge spectrum. Several carrier and envelope periods combine exactly or nearly exactly into observed sidebands.

Period A	Period B	Observed sum period	Mismatch	Bicoherence	Rank A	Rank B	Rank C
57.138	38.535	23.014	0.0000%	0.9226	5	9	15
42.487	47.343	22.093	1.3514%	0.9170	12	13	21
150.636	26.726	23.014	1.3699%	0.9060	4	16	15
414.250	66.280	57.138	0.0000%	0.8744	1	6	5
207.125	25.492	23.014	1.3699%	0.8515	2	17	15
207.125	24.731	22.093	0.0000%	0.8256	2	14	21
38.535	53.452	22.093	1.3514%	0.8101	9	10	21
66.280	118.357	42.487	0.0000%	0.7832	6	11	12
57.138	42.487	24.731	1.4706%	0.7822	5	12	14
57.138	36.022	22.093	0.0000%	0.7735	5	8	21
150.636	57.138	42.487	2.5000%	0.7706	4	5	12
26.726	97.471	21.519	2.5316%	0.7622	16	22	18
150.636	24.731	21.519	1.2821%	0.7567	4	14	18
150.636	25.492	21.519	1.3158%	0.7399	4	17	18
42.487	42.487	21.519	1.2821%	0.7395	12	12	18
414.250	23.014	21.519	1.3158%	0.7286	1	15	18
57.138	47.343	25.492	1.5625%	0.6964	5	13	17
66.280	38.535	24.731	1.4706%	0.6795	6	9	14
118.357	26.726	21.519	1.3158%	0.6775	11	16	18
150.636	66.280	47.343	2.7778%	0.6705	4	6	13
207.125	53.452	42.487	0.0000%	0.6646	2	10	12

Period A	Period B	Observed sum period	Mismatch	Bicoherence	Rank A	Rank B	Rank C
207.125	26.726	23.014	2.8571%	0.6638	2	16	15
414.250	24.731	23.014	1.4085%	0.6606	1	14	15
97.471	97.471	47.343	2.9412%	0.6494	22	22	13
66.280	42.487	25.492	1.5625%	0.6371	6	12	17
66.280	36.022	23.014	1.4085%	0.6328	6	8	15
31.264	66.280	21.519	1.2821%	0.6229	3	6	18
414.250	20.457	19.726	1.1765%	0.6076	1	23	19
36.022	53.452	21.519	0.0000%	0.5987	8	10	18
118.357	97.471	53.452	0.0000%	0.5823	11	22	10
414.250	22.093	21.519	2.5316%	0.5683	1	21	18
207.125	47.343	38.535	0.0000%	0.5683	2	13	9
31.264	118.357	24.731	0.0000%	0.5637	3	11	14
414.250	25.492	24.731	2.8986%	0.5633	1	17	14
47.343	47.343	23.014	2.8571%	0.5458	13	13	15
150.636	38.535	31.264	1.8519%	0.5415	4	9	3
53.452	47.343	25.492	1.5152%	0.5324	10	13	17
66.280	47.343	26.726	3.3333%	0.5288	6	13	16
33.816	118.357	26.726	1.5873%	0.5231	7	11	16
207.125	31.264	26.726	1.6393%	0.5119	2	3	16

15.1 Phase-amplitude coupling

Fast band	Modulation index	Preferred slow phase	Amplitude resultant
18-40	0.02557	+179.67 deg	0.18560
40-80	0.03834	-32.95 deg	0.26658
80-160	0.03541	-45.61 deg	0.27911
160-300	0.02755	+1.58 deg	0.25345

The strongest measured phase-amplitude coupling occurs in the 40-80-turn band, with modulation index approximately 0.0383 and preferred slow phase near -33 degrees. The coupling is moderate but organized. It supports the interpretation that some faster rectilinear carriers are observed through a slower evolving spectral envelope.

16. Angular Harmonic Graph as a Separate Multiplex Layer

The within-turn angular graph is constructed from complete 90-position profiles and is therefore sensitive to orientation around the device circle. The slower rectilinear graph is constructed from temporal evolution along the unwrapped trajectory. These graphs share channels but need not share edges.

The raw profile and the 31-position moving-standard-deviation profile produce different perfect matchings. This rewiring shows that mean angular intensity and angular texture are distinct observables.

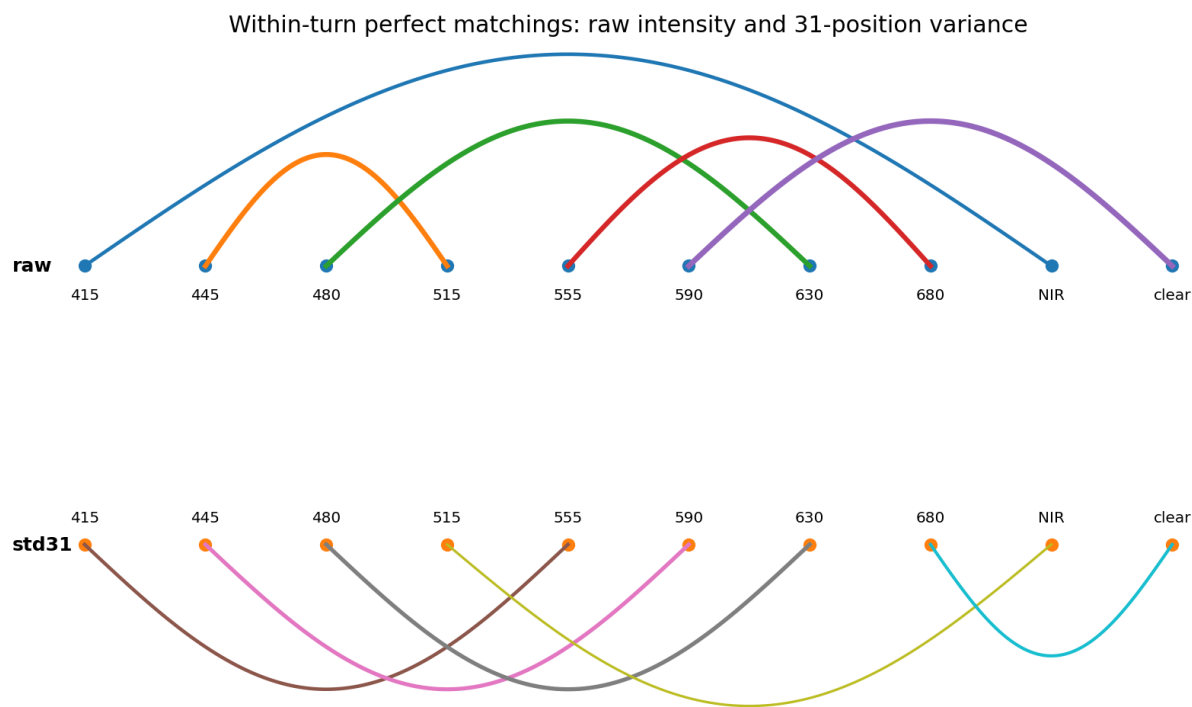


Figure 31. Full-valid angular perfect matchings. The raw layer and the 31-position variance layer share the stable 480-630 pair but rewire the remaining channel graph. Edge thickness follows mean harmonic coherence.

Angular observable	Matched pair	Mean m1-m4 coherence	Phase dispersion
raw	415-NIR	0.2267	2.37 deg
raw	445-515	0.4801	1.30 deg
raw	480-630	0.4880	7.06 deg
raw	555-680	0.4529	2.53 deg
raw	590-clear	0.5643	1.50 deg
std31	415-555	0.2141	8.63 deg
std31	445-590	0.2987	3.56 deg
std31	480-630	0.3154	4.12 deg
std31	515-NIR	0.0322	128.54 deg
std31	680-clear	0.1526	16.52 deg

The raw matching is 415-NIR, 445-515, 480-630, 555-680, and 590-Clear. The 31-position variance matching is 415-555, 445-590, 480-630, 515-NIR, and 680-Clear. Only 480-630 is invariant.

16.1 Direct rectilinear representation of within-turn harmonics

Low-order qA/qB rectilinear bases represent only about 3 to 5 percent of the full complex within-turn harmonic energy. The angular graph and the translation-driven graph are therefore separate multiplex layers. Their strongest bridge is the evolution of the angular-variance state, from which qB is extracted.

Observable	Harmonic	Model	Complex R2	Basis terms
std31	7	AB_lattice	4.62%	27
raw	7	AB_lattice	4.26%	27
std31	5	AB_lattice	4.02%	27
raw	6	AB_lattice	3.68%	27
raw	3	AB_lattice	3.66%	27
raw	5	AB_lattice	3.64%	27
raw	1	AB_lattice	3.62%	27
std31	3	AB_lattice	3.51%	27
std31	4	AB_lattice	3.44%	27
std31	1	AB_lattice	3.38%	27
raw	2	AB_lattice	3.35%	27
std31	6	AB_lattice	3.25%	27
std31	8	AB_lattice	3.22%	27
std31	2	AB_lattice	3.22%	27
raw	4	AB_lattice	3.10%	27
raw	8	AB_lattice	2.68%	27
raw	7	AB_add	2.17%	13
std31	6	AB_add	1.97%	13
raw	3	AB_add	1.92%	13
raw	6	AB_add	1.88%	13
raw	5	AB_add	1.87%	13
std31	5	AB_add	1.87%	13
std31	3	AB_add	1.76%	13
raw	1	AB_add	1.68%	13

17. Regime Continuity and Recovery

The instrument faults change which channels are observable, but the valid segments can still be compared in their common subspaces. Axial cosine between PCA loading vectors measures whether the same channel pattern reappears after the latch interval.

High axial cosine means that a recovered component points along the same channel-state axis, allowing for the arbitrary sign of a principal component.

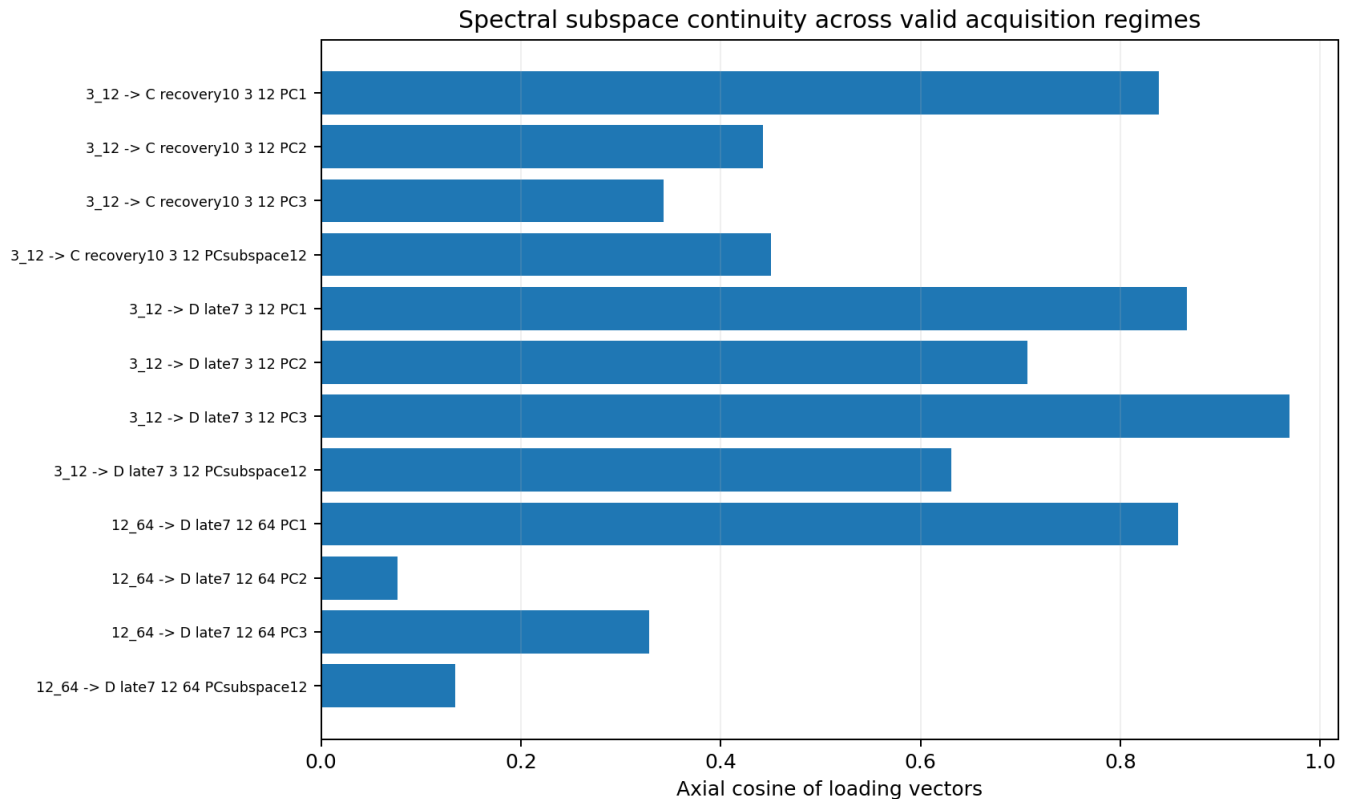


Figure 32. Spectral subspace continuity between the initial full-channel interval and later valid regimes. Some components return along similar loading axes, while others rotate or exchange order as the available channel state changes.

Reference	Comparison	PC	Axial cosine
A_full10_3_12	C_recovery10_3_12	1	0.8383
A_full10_3_12	C_recovery10_3_12	2	0.4420
A_full10_3_12	C_recovery10_3_12	3	0.3429
A_full10_3_12	C_recovery10_3_12	subspace12	0.4501
A_full10_3_12	D_late7_3_12	1	0.8668
A_full10_3_12	D_late7_3_12	2	0.7073
A_full10_3_12	D_late7_3_12	3	0.9695
A_full10_3_12	D_late7_3_12	subspace12	0.6306
A_full10_12_64	D_late7_12_64	1	0.8578
A_full10_12_64	D_late7_12_64	2	0.0763
A_full10_12_64	D_late7_12_64	3	0.3284
A_full10_12_64	D_late7_12_64	subspace12	0.1343

This continuity supports treating the apertures as observations of a related multichannel system rather than unrelated experiments. At the same time, the rank changes prevent a single ten-channel PCA from being extended blindly through the stale intervals.

18. Integrated Geometric Interpretation

The complete record is best represented as a multiplex signal embedded in a rectilinear transverse frame. A broadband common coordinate, a spectral-polarization coordinate, angular-variance coordinates, and nonlinear phase coordinates project differently onto the recovered plane families.

The two-axis frame explains why no single channel or single spacing contains the entire structure. Different detector channels sample different phases and derivatives of the same latent state; different preprocessing observables expose different reciprocal branches; and mixed phases create sidebands and slow torus textures.

18.1 Minimum sufficient structural model

```
x_c(t,theta) ~= C_c s_common(t)
+ P_c s_polarized(Phi_A,Phi_B,t)
+ A_c(t) F_c(theta)
+ sum_(m,n) H_c,mn exp[i 2 pi (m Phi_A + n Phi_B)]
+ epsilon_c(t,theta)
```

C_c and P_c are channel loading vectors. s_{common} and $s_{\text{polarized}}$ are low-dimensional temporal states. $F_c(\theta)$ is the within-turn angular response. $H_{c,mn}$ represents mixed rectilinear harmonics. The model is not asserted as a unique mechanism; it is the minimum descriptive class needed to contain the observed common modes, spectral polarization, angular rewiring, quadrature loops, sidebands, and two-axis phase maps.

18.2 What the exposed base face means

The q_A and q_B plane families define a repeatable transverse address. Advancing by a changes q_A phase by one cycle and leaves q_B phase unchanged. Advancing by b changes q_B phase by one cycle and leaves q_A phase unchanged. The four base nodes therefore define the smallest measured transverse address cell for the selected reciprocal frame.

The cell is open along the CMB-axis direction. The experiment traverses a long distance along that direction, but a nearly constant longitudinal velocity provides insufficient independent geometry to determine a unique longitudinal reciprocal magnitude. The cross product identifies the axis direction with high angular closure; it does not supply the missing period.

18.3 Why small vector errors become large phase errors

The record translates approximately 8.06×10^{10} metres along the supplied CMB direction. A reciprocal-vector error of only 10^{-9} cycles per metre would accumulate approximately 80 cycles. Consequently, the transverse direction can be well organized while global absolute phase remains highly sensitive. This explains why local reciprocal relations and repeated axis families can be visible even when a unique three-dimensional lattice registry is not.

18.4 Paradigm consequence

A rotating multichannel optical probe cannot be interpreted solely as ten independent intensity time series or as a closed angular scan. The information-bearing object is a spatially translated, orientation-resolved, complex multichannel state. Rectilinear structure appears through direction recurrence, derivative-constrained chirps, phase progressions, nonlinear loops, and mixed reciprocal phases. This opens a systematic pathway for extracting spatial organization from signals that look incoherent in any one channel or any one linear projection.

19. Reproduction Procedure

The following procedure reproduces the full-valid exposed-plane analysis from the source CSV and the declared physical constants. Each operation preserves the channel-freshness boundaries and the unwrapped inertial trajectory.

1. Read `pic_id14E_full24_hrs_20260805.csv` and calculate effective UTC as `date.UTC` minus `frame_read_us`.
2. Reconstruct sequential device turns from index-zero resets. Retain only exact 0-through-89 index sequences.
3. Audit channel freshness by first differences, within-turn ranges, and persistence across turns. Establish the four valid apertures before multichannel decomposition.
4. Calculate per-turn channel means, logarithmic means, ranges, and standard deviations.
5. Detrend every 90-position angular profile. Calculate circular 31-position moving standard deviation. Calculate complex harmonics $m = 1$ through 8 for both angular observables.
6. Reconstruct the ICRS trajectory from the supplied CMB velocity, Earth rotation, laboratory coordinates, and device offset. Do not close successive local revolutions.
7. Decompose valid per-turn channel states into the declared temporal bands with zero-phase filtering. Standardize within each aperture and calculate PCA.
8. Calculate analytic signals, complex coherence, synchronized phase, phase-graph eigenstructure, nonlinear dependence, quadrature models, and $m:n$ phase locks.
9. Construct full-aperture time-frequency surfaces. Search geometry-constrained paths $f(t) = |q.v(t)|$ for each selected observable.
10. Cluster candidate reciprocal directions axially. Inspect spacing branches and recurrence across independent observables.
11. Select q_A and q_B from independent spectral-polarization and angular-variance observables. Calculate their cross product, direct dual basis, base-face nodes, and plane equations.
12. Calculate geometric velocity, acceleration, jerk, phase traversal, and projected-velocity reversals for q_A , q_B , q_C , and q_F .
13. Fit low-order one-axis and two-axis Fourier lattice bases. Construct binned ($\text{Phi}_A \bmod 1$, $\text{Phi}_B \bmod 1$) phase maps.
14. Enumerate low-order reciprocal-vector relations, sideband triads, phase-amplitude coupling, and angular perfect matchings.
15. Export the full numerical ledgers, figures, hashes, direct-space nodes, and plane equations to the machine-readable companion.

19.1 Defensive archival requirements

The source hash, every derived ledger hash, generation epoch, coordinate conventions, channel validity rules, base-face node coordinates, and figure manifest are retained in the machine-readable companion. The PDF uses only the core ReportLab fonts Helvetica, Helvetica-Bold, Times-Roman, and Courier. Every table is width constrained to the printable page and every text cell is a wrapping Paragraph.

20. Conclusions

The entire valid pic_id14E signal contains several recurring CMB-transverse axis families rather than one unique plane spacing. Two independently extracted reciprocal directions, qA and qB, form an approximately orthogonal transverse frame whose cross product reconstructs the supplied CMB axis to 0.0364 degrees.

The pair exposes a dimensionally defined direct-space base face. Its direct vectors are 319.393 km and 1,039.778 km long, their included angle is 95.609 degrees, and their area is 330,508.374 square kilometres. The bounding reciprocal plane spacings are 317.864 km and 1,034.800 km. Every base node and plane equation is explicitly listed.

The signal is distributed among a broadband common mode, a spectral-polarization mode, hidden quadrature channel pairs, nontrivial m:n phase locks, angular matching layers, mixed rectilinear phases, carrier-envelope sidebands, and geometric derivative reversals. The slow complex phase graph is more organized than ordinary intensity correlations reveal.

qC is locally related to qA plus 2qB with 1.303 percent vector mismatch. This is a compact reciprocal-space grammar, not global phase closure. The residual accumulates approximately 146 cycles across the full translation, showing that transverse geometry and absolute longitudinal registry are different measurement problems.

The longitudinal direct-space period remains unresolved. The experiment therefore establishes an exposed transverse plane structure and base face embedded in three-dimensional motion, not a closed three-dimensional primitive cell. A future experiment must vary the longitudinal velocity direction or provide an independent third reciprocal phase to close that dimension.

Central archival statement. The measured object is a translated, rotating, complex multichannel state embedded in a two-axis CMB-transverse rectilinear frame. Its information is encoded across phase, scale, channel polarity, nonlinear loops, and reciprocal combinations. The correct analysis space is three-dimensional inertial position plus multiplex spectral and angular state, not a set of independent time series or closed local circles.

Appendix A. Complete Exposed Base-Face Node Patch

The following 4 x 4 patch lists every node displayed in Figure 9. Coordinates are exact linear combinations $i a + j b$ of the recovered direct basis. qA and qB phases are integer by construction.

Node	i	j	ICRS x km	ICRS y km	ICRS z km	qA phase	qB phase
N-1_-1	-1	-1	-344.469340	494.637081	-868.856661	-1.000000	-1.000000
N-1_+0	-1	0	-23.580892	-241.209359	-208.024245	-1.000000	-0.000000
N-1_+1	-1	1	297.307557	-977.055799	452.808171	-1.000000	1.000000
N-1_+2	-1	2	618.196005	-1712.902239	1113.640586	-1.000000	2.000000
N+0_-1	0	-1	-320.888448	735.846440	-660.832416	0.000000	-1.000000
N+0_+0	0	0	0.000000	0.000000	0.000000	0.000000	0.000000
N+0_+1	0	1	320.888448	-735.846440	660.832416	-0.000000	1.000000
N+0_+2	0	2	641.776897	-1471.692880	1321.664831	-0.000000	2.000000
N+1_-1	1	-1	-297.307557	977.055799	-452.808171	1.000000	-1.000000
N+1_+0	1	0	23.580892	241.209359	208.024245	1.000000	0.000000
N+1_+1	1	1	344.469340	-494.637081	868.856661	1.000000	1.000000
N+1_+2	1	2	665.357789	-1230.483521	1529.689076	1.000000	2.000000
N+2_-1	2	-1	-273.726665	1218.265158	-244.783926	2.000000	-1.000000
N+2_+0	2	0	47.161783	482.418718	416.048490	2.000000	0.000000
N+2_+1	2	1	368.050232	-253.427722	1076.880906	2.000000	1.000000
N+2_+2	2	2	688.938680	-989.274162	1737.713321	2.000000	2.000000

Appendix A.1 Plane-family ledger

Family	Index	Offset	Spacing	ICRS plane equation, x/y/z in metres
A	-1	-317.864281 km	317.864281 km	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = -1$
A	0	0.000000 km	317.864281 km	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = 0$
A	1	317.864281 km	317.864281 km	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = 1$
A	2	635.728563 km	317.864281 km	$3.287321093305e-07 x + 2.168679499253e-06 y + 2.255229490922e-06 z = 2$
B	-1	-1034.800487 km	1034.800487 km	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = -1$
B	0	0.000000 km	1034.800487 km	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = 0$
B	1	1034.800487 km	1034.800487 km	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = 1$
B	2	2069.600973 km	1034.800487 km	$3.066748724203e-07 x + -6.155152953415e-07 y + 6.789419322445e-07 z = 2$

Appendix B. Axis-Family Branch Assignments

Family	Observable	Rank	Spacing	Transverse angle	Branch order	Fractional residual	Score
0	bridge_PC2	2	508.231 km	98.133 deg	2.000	+0.0790	0.6487
0	bridge_common	1	364.139 km	85.676 deg	3.000	-0.0983	0.6385
0	bridge_common	2	389.098 km	110.052 deg	3.000	-0.2845	0.5595
0	clean10_PC1	1	260.277 km	102.325 deg	4.000	+0.0595	0.6958
0	clean10_std31_m1_PC1	1	156.744 km	102.717 deg	7.000	-0.2590	1.0414
0	clean10_PC2	5	128.009 km	109.347 deg	8.000	+0.2541	0.7431
1	bridge_differential	5	1373.251 km	36.562 deg	1.000	+0.1008	0.4340
1	bridge_PC2	1	810.147 km	55.169 deg	2.000	-0.1342	0.6722
1	clean10_PC2	6	620.650 km	58.479 deg	2.000	+0.4355	0.6766
1	clean10_PC2	7	689.449 km	54.692 deg	2.000	+0.1925	0.6763
1	clean10_std31_m1_PC2	2	390.945 km	57.834 deg	4.000	-0.1334	0.7241
1	clean10_std31_m1_PC2	1	311.533 km	63.946 deg	5.000	-0.1478	0.7257
2	bridge_differential	7	1251.265 km	119.072 deg	1.000	+0.1900	0.4166
2	bridge_differential	1	670.886 km	128.768 deg	2.000	+0.2195	0.5343
2	bridge_differential	3	768.022 km	131.975 deg	2.000	-0.0613	0.4751
2	clean10_PC2	1	345.577 km	135.904 deg	4.000	+0.3087	0.7602
2	clean10_PC2	3	391.720 km	144.382 deg	4.000	-0.1988	0.7574
2	clean10_PC2	2	317.864 km	131.564 deg	5.000	-0.3156	0.7593
2	clean10_PC2	4	293.779 km	127.530 deg	5.000	+0.0684	0.7566
3	bridge_differential	2	1435.903 km	168.583 deg	1.000	+0.0540	0.4981
3	bridge_differential	4	1589.084 km	11.993 deg	1.000	-0.0476	0.4375
3	bridge_differential	6	1503.967 km	17.575 deg	1.000	+0.0063	0.4284

Appendix C. Synchronized Channel Phase Ledger

Aperture	Band	Channel	Synchronized phase	Loading amplitude	Community
A_full10	12_64	415	+0.000 deg	0.29431	1
A_full10	12_64	445	+3.479 deg	0.32130	1
A_full10	12_64	480	+24.219 deg	0.20821	1
A_full10	12_64	515	-10.784 deg	0.34652	0
A_full10	12_64	555	-16.835 deg	0.34213	0
A_full10	12_64	590	-8.613 deg	0.34892	0
A_full10	12_64	630	-13.886 deg	0.32394	0
A_full10	12_64	680	-14.620 deg	0.34831	0
A_full10	12_64	NIR	-27.326 deg	0.27556	0
A_full10	12_64	clear	-9.089 deg	0.32515	0
A_full10	3_12	415	+0.000 deg	0.30703	1
A_full10	3_12	445	-2.125 deg	0.33520	1
A_full10	3_12	480	-2.140 deg	0.30909	1
A_full10	3_12	515	-6.554 deg	0.31239	0
A_full10	3_12	555	-3.091 deg	0.32685	1
A_full10	3_12	590	-0.929 deg	0.33261	1
A_full10	3_12	630	-2.390 deg	0.31103	1
A_full10	3_12	680	-4.385 deg	0.30675	0
A_full10	3_12	NIR	-2.285 deg	0.30238	1
A_full10	3_12	clear	-2.264 deg	0.31705	1
A_full10	64_300	415	+0.000 deg	0.32049	0
A_full10	64_300	445	-7.465 deg	0.34195	0
A_full10	64_300	480	-2.192 deg	0.30391	0
A_full10	64_300	515	-35.884 deg	0.33983	0
A_full10	64_300	555	-20.932 deg	0.33301	0
A_full10	64_300	590	-21.503 deg	0.34397	0
A_full10	64_300	630	-39.613 deg	0.28667	0
A_full10	64_300	680	-38.975 deg	0.32610	0
A_full10	64_300	NIR	-143.369 deg	0.17840	1
A_full10	64_300	clear	-29.069 deg	0.35065	0
Bridge3	12_64	445	+0.000 deg	0.63034	0
Bridge3	12_64	NIR	-30.708 deg	0.39850	1
Bridge3	12_64	clear	-7.345 deg	0.66624	0
Bridge3	300_1000	445	+0.000 deg	0.62168	0
Bridge3	300_1000	NIR	-119.491 deg	0.53833	1
Bridge3	300_1000	clear	-36.162 deg	0.56896	0
Bridge3	3_12	445	+0.000 deg	0.58822	0
Bridge3	3_12	NIR	+1.253 deg	0.56141	1
Bridge3	3_12	clear	+0.611 deg	0.58208	0
Bridge3	64_300	445	+0.000 deg	0.68144	0
Bridge3	64_300	NIR	-125.963 deg	0.34628	1
Bridge3	64_300	clear	-6.884 deg	0.64478	0
C_recovery10	12_32	415	+0.000 deg	0.32726	0
C_recovery10	12_32	445	+10.604 deg	0.32779	0
C_recovery10	12_32	480	-16.709 deg	0.26529	1
C_recovery10	12_32	515	+13.000 deg	0.33173	0
C_recovery10	12_32	555	+5.654 deg	0.33096	0
C_recovery10	12_32	590	+7.766 deg	0.32404	0
C_recovery10	12_32	630	-27.770 deg	0.28207	1
C_recovery10	12_32	680	+14.552 deg	0.33065	0

Aperture	Band	Channel	Synchronized phase	Loading amplitude	Community
C_recovery10	12_32	NIR	+2.070 deg	0.30839	0
C_recovery10	12_32	clear	+4.962 deg	0.32634	0
C_recovery10	3_12	415	+0.000 deg	0.31298	0
C_recovery10	3_12	445	+5.375 deg	0.36263	0
C_recovery10	3_12	480	+25.859 deg	0.29333	1
C_recovery10	3_12	515	-0.585 deg	0.27148	1
C_recovery10	3_12	555	+6.845 deg	0.34262	0
C_recovery10	3_12	590	+14.436 deg	0.36173	0
C_recovery10	3_12	630	+41.016 deg	0.29269	1
C_recovery10	3_12	680	+17.993 deg	0.23955	1
C_recovery10	3_12	NIR	+29.915 deg	0.33161	0
C_recovery10	3_12	clear	+23.848 deg	0.33098	0
D_late7	12_64	515	+0.000 deg	0.08774	1
D_late7	12_64	555	-101.549 deg	0.41613	0
D_late7	12_64	590	-108.225 deg	0.42735	0
D_late7	12_64	630	-98.225 deg	0.42606	0
D_late7	12_64	680	-94.102 deg	0.41360	0
D_late7	12_64	NIR	-120.377 deg	0.37178	0
D_late7	12_64	clear	-107.518 deg	0.38170	0
D_late7	3_12	515	+0.000 deg	0.07007	1
D_late7	3_12	555	-5.156 deg	0.41183	0
D_late7	3_12	590	-6.668 deg	0.42213	0
D_late7	3_12	630	-5.888 deg	0.40000	0
D_late7	3_12	680	-3.780 deg	0.39802	0
D_late7	3_12	NIR	-9.670 deg	0.39967	0
D_late7	3_12	clear	-5.282 deg	0.41126	0
D_late7	64_180	515	+0.000 deg	0.12100	1
D_late7	64_180	555	+136.641 deg	0.40793	0
D_late7	64_180	590	+145.027 deg	0.40919	0
D_late7	64_180	630	+123.112 deg	0.40139	0
D_late7	64_180	680	+147.239 deg	0.40785	0
D_late7	64_180	NIR	+149.645 deg	0.40071	0
D_late7	64_180	clear	+150.841 deg	0.40435	0

Appendix D. Spectral Phase Polynomial Fits

Aperture	Band	Polynomial degree	Phase-energy R2	Intercept deg	Coefficients
A_full10	12_64	1	0.543071	-55.9521	21.685451648781207
A_full10	12_64	2	0.552003	-80.2054	44.83879737836679;-5.264000480026253
A_full10	12_64	3	0.608262	+209.6128	-390.14318421741694;203.31720659302707;-32.124834959121266
A_full10	3_12	1	0.059818	-4.7372	0.9250880245350698
A_full10	3_12	2	0.251737	+9.7134	-12.870132870524305;3.1363955024810357
A_full10	3_12	3	0.341100	-37.2366	57.59601540008812;-30.653312774979163;5.20415437690934
A_full10	64_300	1	0.708908	-198.1355	72.74290642415909
A_full10	64_300	2	0.909365	-535.4757	394.78318739675024;-73.217072531394
A_full10	64_300	3	0.958696	-1332.2695	1590.6738290609935;-646.666833945865;88.32041582890575
C_recovery10	12_32	1	0.000149	+0.2410	0.3455225967032969
C_recovery10	12_32	2	0.003478	+14.4722	-13.240193826699302;3.0887638706534295
C_recovery10	12_32	3	0.005881	+72.0319	-99.63040387010703;44.51432872748348;-6.380198162569542
C_recovery10	3_12	1	0.400458	+56.1590	-18.00081437270234
C_recovery10	3_12	2	0.401165	+49.5621	-11.703092693898933;-1.4318107770460111
C_recovery10	3_12	3	0.402490	+6.5745	52.8160840173069;-32.36984574670735;4.76495117324641
D_late7	12_64	1	0.437956	-243.7426	79.01227366768249
D_late7	12_64	2	0.674229	+330.1655	-556.748427682813;169.8514907223441
D_late7	12_64	3	0.973533	-5712.5917	9468.04517916036;-5226.66830337629;946.6317921037524
D_late7	3_12	1	0.613605	-18.8537	6.889426050282257
D_late7	3_12	2	0.629817	-7.7795	-5.378307231048826;3.277479688807403
D_late7	3_12	3	0.993896	-498.7275	809.0931332503046;-435.16658436507066;76.90976887483082
D_late7	64_180	1	0.393356	+314.6164	-99.69520722260032
D_late7	64_180	2	0.725766	-591.6891	904.2868375406001;-268.22646728440145
D_late7	64_180	3	0.918937	+5871.5681	-9818.106383644406;5503.823265363934;-1012.5054648190203

Appendix E. Archival and Reproduction Record

Field	Value
Principal Investigator	P. Dwayne Esterline, Principal Investigator
Degrees	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC.
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785932820
Generation UTC	2026-08-05T12:27:00Z
PDF	K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane_structure_1785932820.pdf
JSON	K-Field_spectrophotometer_pic_id14E_analysis_exposed_plane_structure_1785932820.json
PDF engine	Python ReportLab
PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Source record	pic_id14E_full24_hrs_20260805.csv
Source SHA-256	6f5e3c69c623fe02b854e121110f915861bab24e8b6f000b094e810f49b53f62
Analysis companion source	K-Field_spectrophotometer_pic_id14E_fullvalid_exploratory_analysis_1785932633.json
Analysis source SHA-256	7ebd920092387642774c7135d618e3322791202cc0525b4f2cad3d3228ec4b6f
Historical format references	K-Field Spectrophotometer Optimized Geometry; K-Field Spectrophotometer Hidden Structure Energy Analysis; K-Field Spectrophotometer pic_id14C Analysis; K-Field Spectrophotometer Spherical Coherence
Machine-readable scope	Complete exposed base-face geometry, nodes, plane equations, selected reciprocal frame, derived ledgers, hashes, and figure manifest; very large per-turn and angular-harmonic ledgers are represented by exact hashes and row counts.

Appendix E.1 Large-ledger manifest

Filename	Rows	Bytes	SHA-256
angular_harmonics.csv	301,280	25,483,049	35a106927e97068e3602bc8e1126a5cbaacc956c0e1213eb95faf1a9c0fed84e
pic_id14E_fullvalid_pca_scores.csv	9,484	888,923	5c839edd820ef1c3e7875aa50e8215b471b3fd1e15c6829bb8e1bb033a007f5b
pic_id14E_fullvalid_turn_features.csv	1,883	1,611,180	b4dbc165767c4d13fac17a151383abcd2388752c79863599062c71f621d1bd5d

Appendix E.2 Figure manifest

Figure	Bytes	SHA-256
pic_id14E_channel_freshness_regimes.png	71,368	d8f9b68fdc69487434d52b3c938d8b778a4271195e104f84736e0e27f2c079fb
pic_id14E_nonlinear_loop_415_680.png	238,716	80569f87c54605fc2ee4f089381cc9b8c88f45d8469e9ed12cf90d5a5f8868e3
pic_id14E_nonlinear_loop_480_630.png	221,497	e13f71d61f2b30199e04d79a779b034cf7625aca837899deac0cc85fa484bf18
pic_id14E_unwrapped_trajectory_3d.png	355,388	0d068f5ed6918939da7d389ddbca7a20321789f4364aa8b0a1d0b7023aa31c7
pic_id14E_fullvalid_axis_families.png	90,935	a9af66c72adc45d485e023b3055f56d55dd450ccb09cdefe44208130a130bfd0
pic_id14E_fullvalid_bridge_time_frequency_paths.png	329,487	0d5efef70f1f7c772f4a34e9c342784bca8ade8a92db0a5d6f32d28e64d48423
pic_id14E_fullvalid_clean_rectilinear_torus_embedding_3d.png	469,640	a08b2b3c2dddc8a7d0a2280cf5845b4cbe2a481bc89e36bf139ee7dc7976e4bd
pic_id14E_fullvalid_combination_phase_drift.png	69,924	c18be0687a6e973e841d181c5fd93a5bf344215fa96a9ab113bb65603d9388ca
pic_id14E_fullvalid_phase_map_A_full10_64_300_PC1.png	62,778	5933dd2faf21335aada21fa41f1fe573263c15e73e2bcf53d7dbc663a45ddc55
pic_id14E_fullvalid_phase_map_A_full10_64_300_PC2.png	57,394	5838619d85ec9f0c13822ce665b396d1000a664a34405069efc289ba51a8070c
pic_id14E_fullvalid_phase_map_Bridge3_300_1000_PC1.png	62,084	5130f87797ebf106783faad14621f675edfe14f3e9bb87a2dcb2df16797b9e4
pic_id14E_fullvalid_phase_map_Bridge3_300_1000_PC2.png	63,030	f0f63abd3559724059b8ba82c7d3efbef2753e35378680157f138e9b471def23
pic_id14E_fullvalid_qC_reversal_bridge.png	114,771	57f150eb8bc839f5974f7b338ce442e88d81fad9d7c0cdeb0b24398678bb89619
pic_id14E_fullvalid_qC_reversal_late7.png	101,420	7891f7f1feea5f300f34ecc288890179f2de77c03857b13154d282064dcf03e5
pic_id14E_fullvalid_quadrature_gain_ranked.png	172,253	ca3e0bd8f0b7cb95e237c7d50b4823ba99254d346db579718ee094c472b1417
pic_id14E_fullvalid_reciprocal_frame_3d.png	330,518	cfe796863cf8977b4a0dddd573764cf3900fde3635a8aa90ed84036cd317092
pic_id14E_fullvalid_sideband_triads.png	69,308	bc9e1d45a0708750a5e929258021db698e897ca54bfbc997b1486b89fd97ba9
pic_id14E_fullvalid_spectral_phase_progression.png	112,263	afdfc41f2117e9c6383654466c85094bdf2088917e87311eb11f7efa86c0153f
angular_matching_layers.png	150,016	0300608c5d25fb3e2f5fadf0be2551c297b4407ca30688d553de96f3b64a0172
bridge_spectrum_peaks.png	65,816	b2a29cb00e1828663c338a6769b16fe93bc50c71335148bd9cb84979d4fbc096
exposed_plane_families_3d.png	315,298	c93f9e25efe93396ebbaec14f4f54ee16951beb1701aa43c3d5519000160a5c8
exposed_transverse_base_face.png	121,427	45b9b0dd8ac09e5aa0eb177198699c6f9b25cef94a025b7a9a4d154d9b8df20e
geometric_velocity_components.png	52,843	892a006870817159ec054fcb780e3dc746a5a6e8704ffc55b9968692815a58e3
late_phase_locks_2to1.png	88,942	da32e67cc3f7f98ef294a2c126b5796027a1e5913643815cf061e4413ace53aa
multiscale_participation_dimension.png	123,791	b52fc03e91c4b6e1d9f1a7038b2fd17fc0fa692f243bc889813af6b9b825da82
named_path_score_matrix.png	145,668	b60fb51ad192f2c9010e9ebab2b79975e5dec49d7eb9397534f889aee6333e0
pca_loadings_12_64.png	51,686	bdb5657efb40eab3283158bc1b65e838c00c1ed80dff430b05d1bbaa8c0d4d25
pca_loadings_64_300.png	52,829	d8a9fc8a52455ddf86b76bb7971e619c26cbabe533f9b225540a31741c4ed2a6
phase_graph_dimension.png	146,560	85ed0714ab34c3a1edb54f4340b539ee19530a020df1f309d4b73842ad08a92f
reciprocal_vector_closure.png	102,541	b96bba896d81ed06e89bb2e2561543843500d4b48fc196908ee3dfc854747413
spatial_phase_traversal.png	56,369	503245df45a6d9af4fe5fa39314a47a634d0647f6e648bd455829b97d9445227
subspace_continuity.png	96,281	43352a42f7bc37fbee6bf45a3ef15a5e3f30514bf3dc698acd992e23a793b364
title_exposed_structure_composite.png	311,926	a199ca4f3a53196737a9cf4d80afc7557003cc91266d14241148435f2f5bf569

END OF PROPRIETARY CONFIDENTIAL TRADE SECRET ARCHIVAL REFERENCE